

Master Thesis

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Topic:

Scalable Multi-Agent Reinforcement Learning for Adaptive Traffic Signal Control in Urban Environments



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Author Declaration

I hereby declare that this thesis is my own work and that I have clearly acknowledged all sources used in its creation. No portion of the work referred to in this thesis has been submitted in support of an application for another degree or qualification at this or any other university or institute of learning.

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Date: _____

Abstract

Urban traffic signal control is a networked sequential decision problem in which local phase decisions influence queues, downstream arrivals and future congestion. This thesis designs and evaluates a scalable independent Multi-Agent Reinforcement Learning (MARL) framework for adaptive traffic signal control using Python, Simulation of Urban MObility (SUMO) and the Traffic Control Interface (TraCI). Each signalised intersection is modelled as an independent agent trained with either Deep Q-Networks (DQN) or Proximal Policy Optimisation (PPO). The final implementation uses a validated HOLD/SWITCH action model with a 10-second minimum green, a three-second yellow transition and a five-second decision interval.

The final evaluation contains 175 unique deterministic runs and compares five controllers: Original Fixed-Time, Webster-derived Fixed-Time, Actuated, DQN and PPO. Demand sensitivity is evaluated on a 2×2 network under low, medium, high and dynamic traffic, while scalability is tested under calibrated medium demand on 2×2 , 3×3 , 4×4 and 5×5 networks containing 4–25 controlled intersections. Statistical analysis uses matched evaluation seeds, paired tests, Holm–Bonferroni multiplicity correction, confidence intervals and paired effect sizes.

Results show that baseline strength materially changes the interpretation of MARL performance. DQN and PPO substantially outperform the weak Original Fixed-Time controller under several low, medium and dynamic conditions, but Actuated control is the strongest descriptive medium-demand controller across all tested grid sizes. On the 2×2 medium network, DQN is statistically indistinguishable from Actuated across the five primary traffic metrics. From 3×3 onward, DQN preserves a statistically similar completion rate but develops significant travel-time, waiting-time and queue disadvantages. PPO scales less favourably and deteriorates substantially in the severe high-demand stress test, where even the best controller completes only 15.53% of planned trips.

Computationally, independent MARL remains lightweight at inference time within the tested 4–25-agent range. At 25 agents, mean joint decision latency is approximately 1.04 ms for DQN and 1.57 ms for PPO, both below 0.032% of the five-second control budget, although training time, model storage and DQN replay

memory increase with agent count.

The thesis concludes that independent DQN is a viable research approach with computationally lightweight inference across the tested 4–25-agent range, but the evidence does not support universal MARL superiority over strong conventional adaptive control. Robust evaluation requires strong baselines, multiple demand regimes, reproducible statistical analysis and explicit separation of architectural, behavioural and computational scalability.

Keywords: adaptive traffic signal control; multi-agent reinforcement learning; DQN; PPO; SUMO; TraCI; scalability; actuated control; intelligent transportation systems

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List of Abbreviations

A2C	Advantage Actor–Critic
ACM	Association for Computing Machinery
ATSC	Adaptive Traffic Signal Control
CPU	Central Processing Unit
CTDE	Centralised Training with Decentralised Execution
DDQN	Double Deep Q-Network
DOI	Digital Object Identifier
DQN	Deep Q-Network
DRL	Deep Reinforcement Learning
FHWA	Federal Highway Administration
GAE	Generalised Advantage Estimation
GPU	Graphics Processing Unit
IEEE	Institute of Electrical and Electronics Engineers
MARL	Multi-Agent Reinforcement Learning
MDP	Markov Decision Process
MOVA	Microprocessor Optimised Vehicle Actuation
PPO	Proximal Policy Optimisation
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RAM	Random Access Memory
RL	Reinforcement Learning
SHA-256	Secure Hash Algorithm 256-bit
SUMO	Simulation of Urban MObility
SurTrac	Scalable Urban Traffic Control
TraCI	Traffic Control Interface

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Chapter 1 Introduction

1.1 Introduction

Urban road networks operate under traffic demand that varies by time, direction and location. When demand exceeds the effective capacity of a road network, congestion increases travel delay and queueing, which motivates adaptive traffic-signal research at both isolated intersections and network scale (Noaeen et al. 2022).

Traffic signal control is one of the principal operational mechanisms available for regulating vehicle movement through urban intersections. Conventional fixed-time control applies pre-defined phase sequences and green durations, while actuated control changes timing in response to detected traffic conditions. These approaches remain important because they are interpretable, stable and operationally established. However, rapidly changing arrivals and interactions between neighbouring intersections create a networked decision problem in which locally appropriate actions can influence downstream queues and future arrivals.

Reinforcement Learning (RL) provides a data-driven alternative in which an agent learns a control policy by interacting with an environment and receiving feedback through a reward signal. In a traffic-signal network, Multi-Agent Reinforcement Learning (MARL) extends this idea by representing several controlled intersections as interacting decision-making agents. Network-level RL traffic-signal control has consequently received substantial research attention, although reviews continue to identify large variation in network design, simulation environment, controller formulation, evaluation practice and reproducibility (Noaeen et al. 2022).

This thesis investigates a deliberately simple independent-agent MARL architecture. Each signalised intersection is controlled by its own Deep Q-Network (DQN) or Proximal Policy Optimisation (PPO) model within the SUMO microscopic traffic simulator. The work does not assume that a learned controller must outperform conventional traffic engineering. Instead, the final experiment compares learned control with three increasingly strong conventional references: an Original Fixed-Time programme, a Webster-derived Fixed-Time programme and an Actuated controller. This stronger comparison is important because RL traffic-signal studies can produce

very different conclusions depending on the quality of the baseline against which the learned controller is evaluated (Cabrejas-Egea et al. 2021).

The study also treats scalability as a measurable property rather than an architectural claim. The same independent-agent framework is tested on synthetic networks containing 4, 9, 16 and 25 controlled intersections. Traffic performance is evaluated together with training runtime, memory consumption, model size and deterministic inference latency. The final evidence therefore distinguishes architectural scalability, behavioural scalability and computational scalability.

1.2 Background of the Study

Traffic-signal control is inherently sequential. A decision to hold or change a signal phase affects the vehicles discharged during the current interval, the queues that remain, the traffic arriving at neighbouring intersections and the state on which the next decision will be made. This temporal dependence makes RL a natural modelling framework.

A single centralised RL controller could theoretically observe and control an entire network. In practice, the joint state and action space grows quickly with network size, which is a central motivation for decentralised MARL traffic control (Chu et al. 2020). MARL distributes this decision process across multiple agents. Each intersection can make a local decision using local and limited neighbouring information while the combined behaviour of all agents determines network-level traffic performance.

Independent MARL is an especially simple decentralised formulation. It avoids a shared critic, global action model and parameter-sharing mechanism. This simplicity reduces implementation complexity and allows the number of agents to increase without changing each agent’s observation or action dimensionality. The trade-off is that independent agents have incomplete information about the wider network and face partial-observability and coordination limitations (Chu et al. 2020, Noeen et al. 2022).

The present study uses this trade-off as the central empirical question. Rather than introducing a highly complex coordination mechanism, it establishes how far a compact independent-agent architecture can scale before traffic-efficiency disadvantages become visible relative to strong conventional control.

1.3 Problem Statement

Existing research has demonstrated that RL and MARL can improve traffic-signal operation in simulation. However, three methodological problems make those improvements difficult to interpret consistently.

First, baseline strength varies substantially. A learned controller may show a large advantage over a poorly calibrated cyclic controller while showing little or no advantage over an actuated or commercially established controller. Cabrejas-Egea et al. (2021) specifically highlighted the need to compare RL approaches with stronger operational traffic-control systems.

Second, traffic-demand conditions vary across studies. A controller that performs well under light or moderate demand may behave differently as queues approach saturation. Conclusions based on a single demand regime can therefore obscure important failure conditions.

Third, scalability is often discussed without measuring all of its dimensions. A program may successfully instantiate more agents while traffic performance degrades or computational requirements become impractical. Architectural generalisation alone is therefore insufficient evidence of scalable control.

The research problem addressed in this thesis is consequently the lack of a consistent, reproducible and sufficiently demanding evaluation of a simple independent-agent MARL traffic-signal controller across multiple demand regimes, stronger conventional baselines and increasing network sizes. The study addresses this problem using a common simulation environment, fixed performance definitions, matched evaluation seeds, multiplicity-adjusted statistical analysis and computational-resource measurement.

1.4 Research Aim

The aim of this study is to design, implement and critically evaluate a scalable independent Multi-Agent Reinforcement Learning framework for adaptive traffic signal control using Python and SUMO, and to compare DQN- and PPO-based controllers with conventional fixed-time and actuated control across varying traffic demand and network scale. The focus on decentralised control, strong baselines and explicit network-scale evaluation is motivated by prior work showing that traffic-signal RL conclusions depend strongly on coordination architecture, network size and comparator strength (Chu et al. 2020, Noaen et al. 2022, Cabrejas-Egea et al. 2021, Chen et al. 2020).

1.5 Research Objectives

The study has the following objectives, operationalising evaluation dimensions repeatedly identified in the traffic-signal RL literature: controller design, demand sensitivity, baseline quality, network scale, reproducibility and operational performance measurement (Noaeen et al. 2022, Wei et al. 2021, Chen et al. 2020):

1. To design a generalised Python–SUMO MARL architecture in which signalised intersections and direct neighbours are discovered automatically from the traffic network.
2. To implement and train independent DQN and PPO traffic-signal agents using a consistent observation, action and reward formulation.
3. To evaluate learned and conventional controllers under low, medium, high and dynamic demand on a primary 2×2 network.
4. To compare MARL with Original Fixed-Time, Webster-derived Fixed-Time and Actuated control using trip completion, travel time, waiting time, time loss and queue length.
5. To evaluate behavioural scalability on 2×2 , 3×3 , 4×4 and 5×5 synthetic networks containing 4–25 controlled intersections.
6. To measure computational scalability using training runtime, memory consumption, model storage and deterministic joint decision latency.
7. To apply a reproducible Data Science workflow using matched evaluation seeds, paired statistical comparison, Holm–Bonferroni multiplicity correction, effect sizes, confidence intervals and frozen final evidence.

1.6 Research Questions

The following research questions guide the thesis:

- RQ1:** How can an adaptive urban traffic signal control system based on a scalable independent MARL approach be designed and implemented using Python and SUMO?
- RQ2:** How effective are independent DQN and PPO traffic-signal controllers under low, medium, high and dynamic traffic-demand conditions?

RQ3: How well does MARL-based control improve trip completion, waiting time, queue length, travel time and time loss compared with conventional fixed-time and actuated traffic signal control?

RQ4: How can a reproducible Data Science workflow be used to assess the behavioural and computational scalability of the MARL framework as the number of controlled intersections increases?

RQ1 and RQ4 address implementation and scalability, whereas RQ2 and RQ3 address demand sensitivity and controller effectiveness. This separation follows large-scale traffic-signal research in distinguishing controller architecture from operational performance and computational feasibility (Chen et al. 2020, Xing et al. 2022, Noaeen et al. 2022).

1.7 Research Justification

The research is justified at both an application and methodological level.

At the application level, traffic-signal control can influence congestion without requiring immediate physical expansion of road infrastructure. Adaptive control is especially relevant when demand changes over short time periods and traffic conditions at neighbouring intersections are interdependent.

At the methodological level, MARL traffic-signal research needs comparisons that do not rely on weak baselines or a single favourable traffic regime. The systematic review by Noaeen et al. (2022) demonstrates the breadth of network-level RL traffic-signal research while also showing substantial heterogeneity in experimental settings and evaluation. A reproducible experiment that includes multiple conventional controllers, several demand regimes and explicit computational measurements can therefore contribute useful evidence even if the learned controllers do not dominate every baseline.

The study further contributes by reporting unfavourable results rather than selecting only scenarios in which MARL performs well. The severe high-demand scenario, for example, is retained as an oversaturated stress test even though PPO performs poorly. This strengthens the interpretability of the conclusions.

1.8 Scope of the Study

The scope is intentionally bounded. The study uses microscopic simulation in SUMO and does not include field deployment (Alvarez Lopez et al. 2018). Networks are synthetic grids rather than replicas of real cities; this supports controlled scaling

experiments but does not reproduce the geometric and demand heterogeneity of operational networks (Zhang et al. 2019, Bie et al. 2024).

The primary demand experiment uses a 2×2 network with low, medium, high and dynamic traffic conditions. Scalability is tested under calibrated medium demand on networks from 2×2 to 5×5 . The learned controllers are independent DQN and independent PPO. The thesis does not implement centralised training with decentralised execution, graph neural networks, parameter sharing or explicit message-passing between agents. These are established alternatives for increasing coordination in multi-intersection traffic control (Kuyer et al. 2008, Wei, Xu, Zhang, Zheng, Zang, Chen, Zhang, Zhu, Xu & Li 2019, Chu et al. 2020, Liu et al. 2023).

Traffic performance is assessed using trip completion rate, average travel time, average waiting time, average time loss and mean sampled queue length. Teleports are retained as a simulation-integrity diagnostic. Emissions, pedestrian delay, public-transport priority, safety and fairness are outside the final objective; recent safety- and deployment-oriented studies demonstrate that these objectives require explicit treatment rather than being inferred from delay performance alone (Du et al. 2023, Gu et al. 2024).

Computational scalability is measured only on the hardware and software environment used for this thesis and over the tested 4–25-agent range. The study therefore does not claim universal runtime, memory or city-scale complexity characteristics (Xing et al. 2022).

1.9 Significance of the Study

The significance of the research lies in the combination of implementation, critical comparison and reproducible evaluation.

Technically, the project demonstrates that the same independent-agent control code can operate across networks containing four to twenty-five controlled intersections while keeping each agent’s local observation and action space fixed. Large-scale studies show why this architectural property matters while also demonstrating that coordination and data requirements change materially as network size increases (Chen et al. 2020, Liu et al. 2023).

Empirically, the study shows that the conclusion changes when stronger conventional baselines are introduced. Large improvements relative to the Original Fixed-Time controller do not imply superiority over Actuated control, reinforcing the importance of comparator strength in RL traffic-signal evaluation (Cabrejas-Egea et al. 2021).

Analytically, the study separates descriptive performance, statistical evidence

and computational feasibility. Matched seeds, Holm-adjusted paired comparisons, confidence intervals and effect sizes reduce the risk that small or selectively chosen differences are over-interpreted. Reproducibility and evaluation consistency remain recognised concerns in traffic-signal RL research (Noaeen et al. 2022, Wei et al. 2021).

Finally, the study provides a reproducible experimental framework that can be extended to more heterogeneous networks, additional training seeds, cooperative MARL methods, noisy observations and more realistic deployment constraints.

1.10 Structure of the Thesis

The remainder of the thesis is organised to follow the logic of empirical traffic-signal RL research: establish the evidence base, define the simulator and controller design, report operational and statistical results, interpret them against prior work, and delimit the resulting claims (Noaeen et al. 2022, Wei et al. 2021).

Chapter 2 reviews conventional and adaptive traffic-signal control, RL and MARL foundations, previous network-level traffic-signal studies, Data Science evaluation practices, scalability and the research gap.

Chapter 3 describes the research philosophy, simulation environment, corrected signal-control semantics, independent-agent architecture, state/action/reward formulation, learning algorithms, three conventional baselines, demand calibration, final 175-run experimental design, statistical analysis and computational measurements.

Chapter 4 reports the final demand-sensitivity, medium-demand scalability, statistical and computational results.

Chapter 5 interprets those findings against the research questions and verified literature, with particular attention to baseline strength, oversaturation, DQN–PPO differences and the distinction between architectural, behavioural and computational scalability.

Chapter 6 consolidates the answers to the research questions, states the principal contributions and limitations, and presents recommendations for future research.

1.11 Chapter Summary

This chapter introduced adaptive traffic signal control as a networked sequential decision problem and positioned independent MARL as a scalable but coordination-limited approach. The research problem was defined around the need for stronger baselines, multiple traffic-demand regimes, explicit scalability testing and reproducible analysis. This positioning is consistent with literature that treats decentralisation as a mechanism for network-scale control while recognising coordination, evaluation

and deployment limitations (Chu et al. 2020, Chen et al. 2020, Noaeen et al. 2022). The chapter then established the research aim, seven objectives and four research questions that structure the remainder of the thesis. Chapter 2 now examines the literature needed to evaluate these questions critically.

Chapter 2 Literature Review

2.1 Introduction

Adaptive Traffic Signal Control (ATSC) has evolved from pre-timed and detector-responsive engineering methods toward data-driven approaches that learn control policies from traffic observations. Reinforcement Learning (RL) is attractive in this context because signal control is a sequential decision problem: the value of a phase decision depends not only on the vehicles released immediately, but also on how that decision changes downstream queues, future arrivals and later control opportunities. Multi-Agent Reinforcement Learning (MARL) extends this formulation to networks in which several signalised intersections must act simultaneously.

The literature, however, does not support a simple conclusion that MARL is always superior to conventional traffic engineering. Network structure, demand intensity, state representation, reward design, coordination mechanism, baseline strength, simulation configuration and evaluation protocol all influence reported results. The systematic review by Noaen et al. (2022), which analysed 160 peer-reviewed network-level RL traffic-signal studies published between 1994 and March 2020, demonstrates the breadth of the field while also documenting considerable methodological heterogeneity.

This chapter therefore takes a critical rather than purely descriptive approach. It first documents the structured literature-search process used for the revised thesis. It then reviews conventional traffic-signal control, the RL and MARL foundations relevant to this research, state/action/reward design, coordination and scalability, simulation and reproducibility, baseline selection and recent developments in interpretable and cooperative traffic control. The final sections synthesise these strands into the specific research gap addressed by the experimental design in Chapters 3–6.

2.2 Structured Literature Search Strategy

2.2.1 Purpose and Scope

A structured literature search was carried out for the final revision of this thesis to ensure that the review was transparent, reproducible and based on legitimate publications. The process is not presented as a new Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) systematic review. Instead, it is a documented evidence-identification procedure appropriate to the scope of a Master’s dissertation. The systematic review by Noaen et al. (2022) is used as the principal historical synthesis, while the structured search extends the discussion with later primary studies and methodologically important earlier work.

The principal publication window for traffic-signal-control research was 2019 to August 2026. Earlier sources were retained only where they provide foundational methods or software directly used in the thesis, including DQN, PPO and SUMO.

2.2.2 Information Sources and Search Terms

The search focused on peer-reviewed literature and authoritative technical sources accessible through publisher or institutional indexes. The principal sources were the Association for Computing Machinery (ACM) Digital Library, Institute of Electrical and Electronics Engineers (IEEE) Xplore, ScienceDirect/Elsevier, publisher-indexed conference and journal records, the official SUMO publication catalogue and official traffic-signal engineering documentation. Foundational RL algorithm papers were verified through the original publisher or author-hosted research record.

The main search strings combined the following concepts:

- “traffic signal control” AND “reinforcement learning”;
- “multi-agent reinforcement learning” AND “traffic signal control”;
- “large-scale” OR “scalability” AND “traffic signal control” AND “reinforcement learning”;
- “cooperative” AND “multi-intersection” AND “traffic signal control”;
- “actuated” OR “commercial systems” AND “reinforcement learning” AND “traffic signal control”;
- “state reward design” AND “reinforcement learning” AND “traffic signal control”;
- “interpretability” AND “multi-agent reinforcement learning” AND “traffic signal control”;

- “SUMO” AND “multi-agent” AND “traffic signal control”.

2.2.3 Eligibility and Screening

A publication was included in the core synthesis when it satisfied the following criteria: it addressed traffic-signal control; used RL, deep RL or MARL; reported a quantitative evaluation; and contributed directly to at least one thesis theme such as coordination, network scale, state/reward formulation, baseline quality, reproducibility, interpretability or deployment realism. Network-level studies with two or more intersections were prioritised. Single-intersection studies were retained only when they contributed a methodological point directly relevant to the thesis, such as state/reward consistency.

Publications were excluded when the available bibliographic information could not be independently verified, when the work did not concern traffic-signal control, when it was purely conceptual without an empirical evaluation, or when its contribution was outside the thesis scope. Preprints were avoided when a peer-reviewed version was available.

For every source retained in the final evidence matrix, title, authorship, publication venue, year and Digital Object Identifier (DOI) or official publication record were checked. This step is particularly important because an unverified citation can undermine both the reproducibility and academic integrity of a literature review.

2.2.4 Search Protocol and Screening Record

A countable final screening pass was completed on 17 August 2026 so that the revised review could report an auditable selection flow rather than reconstructing numbers from memory. Eight query families derived from the search concepts in Section 2.2.2 were executed against official publisher-indexed records spanning ACM, AAAI, IJCAI, Springer, IEEE and ScienceDirect/Elsevier. The final pass returned 68 records. Four duplicate publication records (for example, HTML and PDF records of the same paper) were removed, leaving 64 unique records for title/abstract screening. Twenty-eight records were excluded at that stage because they were outside the thesis scope, single-intersection studies without a directly relevant methodological contribution, secondary reviews duplicating the principal synthesis, or records whose contribution did not address the defined themes. Thirty-six full texts or complete official publisher records were then assessed; twelve were excluded after detailed relevance checking, leaving 24 records retained from the counted final pass. Earlier foundational and already-verified studies remained eligible where directly required for theory, algorithms, simulation software or comparison context.

Figure 2.1 reports this screening flow. The procedure is described as a structured, countable evidence-identification process rather than a registered PRISMA systematic review; the purpose of the counts is transparency and reproducibility, not to claim exhaustive coverage of every publication in the field.

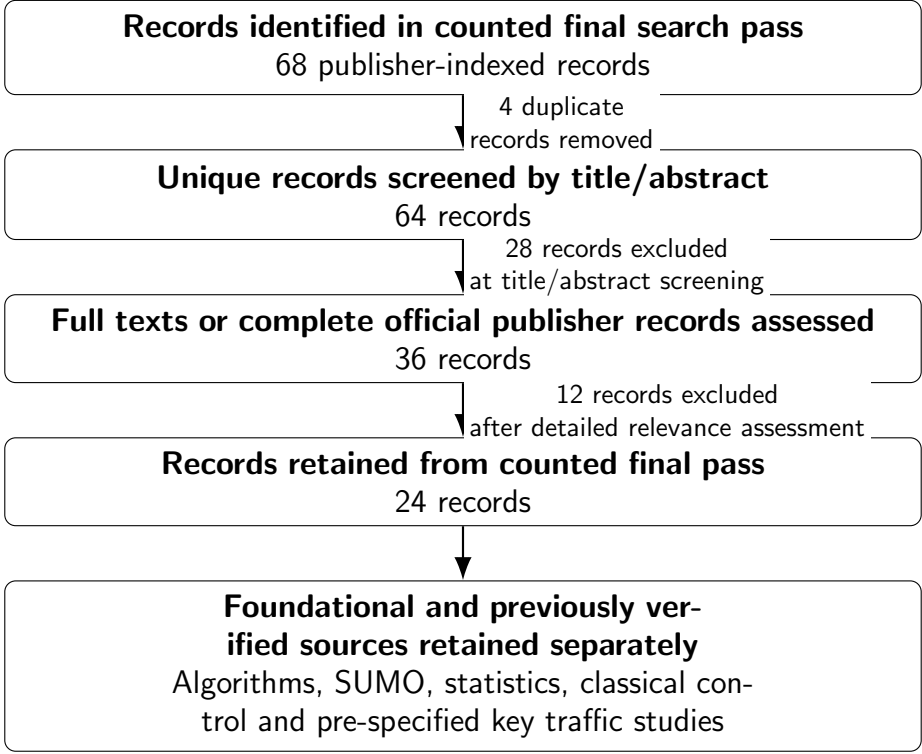


Figure 2.1: Counted literature identification and screening flow used for the final thesis revision.

Source: Author’s retained screening log for the final revision (17 August 2026).

Table 2.1: Structured literature-search and source-verification protocol used for the final revision.

Source	Primary use	Verification retained in the review
ACM Digital Library	Network-level RL/MARL conference studies	Publication title, authors, venue and DOI checked against the publisher record.
IEEE Xplore / official IEEE records	Large-scale traffic control and SUMO-related publications	Publication metadata and DOI/official conference record checked.
ScienceDirect / Elsevier	Traffic-signal RL, MARL, scalability and interpretability studies	Journal, volume/article number, authors and DOI checked.
Official SUMO documentation	Simulator definition and recommended SUMO citation	Current official SUMO publication recommendation checked.
Federal Highway Administration (FHWA) / transportation institutional records	Conventional actuated-control principles	Official report title and publication number checked.
AAAI / IJCAI / Springer publisher records	Recent large-scale, resource-aware, safe and interpretable traffic-signal RL	Complete publisher metadata and DOI checked before retention.

Source: Author’s documented literature-verification procedure for the final thesis revision.

2.2.5 Evidence Extraction

The selected papers were examined using a common evidence framework: controller type; number or type of intersections; coordination mechanism; state/reward concept; simulator or real-network basis; principal comparison baseline; evaluation metrics; and relevance to the present research gap. This approach prevents the review from becoming a sequence of isolated article summaries and instead allows recurring methodological tensions to be compared across studies.

For abbreviation clarity, the principal learning architectures discussed in the evidence matrix are introduced here in full: Deep Reinforcement Learning (DRL), Double Deep Q-Network (DDQN), Advantage Actor–Critic (A2C), and Centralised Training with Decentralised Execution (CTDE). These terms are used in abbreviated form thereafter. The conventional systems Microprocessor Optimised Vehicle Actuation (MOVA) and Scalable Urban Traffic Control (SurTrac) are likewise introduced

in full before their abbreviated use in the evidence matrix.

Table 2.2 summarises the core studies used repeatedly in this chapter.

Table 2.2: Core evidence matrix for the structured literature synthesis.

Study	Control approach	ap-	Network / emphasis	Main methodological contribution	Relevance to this thesis
Kuyer et al. (2008)	Coordinated MARL		Urban traffic network	Coordination graphs and max-plus message passing	Establishes early explicit neighbour coordination as an alternative to independent learning
Wei et al. (2018)	Deep RL		Real traffic data	Traffic-specific state/reward design for adaptive signal control	Demonstrates the importance of domain-aware representation design
Wei, Chen, Zheng, Wu, Gayah, Xu & Li (2019)	RL / max-pressure-inspired		Arterial network	Links state and reward design to max-pressure traffic theory	Motivates interpretable, traffic-grounded reward design
Zheng et al. (2019)	Deep RL		Multiple intersection configurations	Phase-competition representation	Shows that action/phase structure can embed traffic-domain inductive bias
Zhang et al. (2019)	Simulation environment		Large city traffic scenarios	Efficient multi-agent traffic simulation	Provides large-scale simulation context beyond SUMO-based experiments
Wei, Xu, Zhang, Zheng, Zang, Chen, Zhang, Zhu, Xu & Li (2019)	Cooperative MARL		Large network	Graph-attention communication among neighbouring signals	Demonstrates explicit coordination as an alternative to independent agents
Chu et al. (2020)	Decentralised multi-agent A2C		Large synthetic and Monaco networks	Scalable decentralised control under partial observability	Directly motivates scalability and decentralisation analysis
Chen et al. (2020)	Decentralised DRL		Up to 2,510 traffic lights	Pressure-based implicit coordination at large scale	Shows that scalability, coordination and data feasibility are linked design problems
Cabrejas-Egea et al. (2021)	DQN / actor-critic		Variable map scenarios	Comparison against MOVA, SurTrac and cyclic control	Motivates strong baseline selection
Wei et al. (2021)	Survey		RL traffic-signal evaluation	Synthesises models, states, rewards and evaluation practices	Supports systematic interpretation of design and evaluation choices
Haddad et al. (2022)	Cooperative DQN MARL		Multiple intersections	Neighbour state/action/reward information enters learning	Contrasts explicit cooperation with independent learning
Wu et al. (2022)	Distributed DRL		Large-scale traffic-signal control	Distributed RL and computing architecture	Supports separate computational-scalability analysis
Noaen et al. (2022)	Systematic review		160 network-level studies	Synthesises methods, environments, code availability and evaluation	Establishes field-wide heterogeneity and reproducibility concerns
Xing et al. (2022)	Resource-aware DRL		Multiple real-demand networks	Explicit memory/computation optimisation for edge devices	Shows computational cost is part of deployment scalability

Table 2.2 continued

Study	Control proach	ap-	Network / emphasis	Main methodological contribution	Relevance to this thesis
Bouktif et al. (2023)	DDQN		State/reward design	Consistent and interpretable state-reward formulation	Supports explicit analysis of observation and reward design
Du et al. (2023)	Safety-enhanced RL		Synthetic and real benchmarks	Adds explicit safety constraints and multi-objective learning	Shows delay optimisation alone is insufficient for deployment claims
Liu et al. (2023)	Grouped MARL		Up to 1,089 intersections	Dynamic grouping and parameter sharing by agent similarity	Provides a recent large-scale complexity/coordination alternative
Bie et al. (2024)	Value-decomposition MARL		Heterogeneous intersections	Spatiotemporal graph attention and heterogeneity-aware control	Shows regular synthetic grids are a limited scalability test
Song et al. (2024)	Counterfactual actor-critic		Real-world-based Shenzhen network	Centralised critic plus scheduler-assisted communication	Illustrates CTDE-style coordination beyond independent agents
Gu et al. (2024)	Programmatic interpretable RL		Cross-city / resource-limited	Interpretable program policies for constrained deployment	Links interpretability, generalisation and edge deployment
Cai et al. (2025)	Interpretable MARL		Synthetic and real networks	Regulatable policies and interpretable reward construction	Highlights interpretability as a deployment requirement
Ren et al. (2025)	Cooperative MARL		Network-wide traffic-signal control	Tests conditions under which independent control can outperform co-operation	Reinforces that cooperation is not automatically beneficial
Katzilieris et al. (2026)	MARL CTDE	/	Urban arterial, multimodal	Joint signal and dynamic bus-lane control	Shows recent expansion toward integrated multimodal control

Source: Author’s structured synthesis of verified publisher and institutional publication records.

2.3 Evolution of Scalable and Deployment-Oriented RL Traffic Control

2.3.1 Traffic-Specific Representation and Control Design

Early deep-RL traffic-signal systems demonstrated that controller performance depends not only on the learning algorithm but also on how traffic structure is represented. IntelliLight uses traffic-specific state and reward design on real traffic data (Wei et al. 2018), while FRAP models competition between signal phases rather than treating actions as unrelated categorical outputs (Zheng et al. 2019). PressLight similarly grounds learning in max-pressure traffic theory (Wei, Chen, Zheng, Wu, Gayah, Xu & Li 2019). Together, these studies support the present thesis decision to document observation variables, HOLD/SWITCH semantics and reward components explicitly instead of treating the environment interface as an implementation detail.

2.3.2 Coordination and Network Scale

Coordination has been a central MARL traffic-control problem for more than a decade. Kuyer et al. (2008) used coordination graphs and max-plus message passing between neighbouring traffic lights, demonstrating that explicit coordination can improve behaviour when local actions are interdependent. Later work such as CoLight introduced graph-attention communication (Wei, Xu, Zhang, Zheng, Zang, Chen, Zhang, Zhu, Xu & Li 2019), while Chen et al. (2020) showed that decentralised RL can be evaluated at the scale of thousands of traffic lights and framed scalability, coordination and data feasibility as joint challenges. GPLight addresses the complexity of large networks by dynamically grouping agents and sharing parameters within groups, reporting experiments with up to 1,089 intersections (Liu et al. 2023).

These methods are deliberately more coordinated than the independent architecture tested here. Their relevance is therefore comparative: they show what information-sharing or parameter-sharing mechanisms may be required if the performance gap of independent agents grows with network size.

2.3.3 Simulation, Resource Cost and Deployment Constraints

Large-scale experimentation also depends on simulation and computing infrastructure. CityFlow was developed specifically as an efficient multi-agent environment for large city traffic scenarios (Zhang et al. 2019), illustrating that simulator throughput becomes part of the research design as networks expand. TinyLight moves the scalability question closer to deployment by explicitly optimising storage and computation for highly resource-constrained devices (Xing et al. 2022). These studies motivate the present thesis distinction between behavioural scaling and computational scaling.

Deployment-oriented work also demonstrates that delay and queue metrics are not sufficient on their own. SafeLight incorporates explicit traffic-safety objectives into RL traffic control (Du et al. 2023), while π -Light uses programmatic policies to improve interpretability, generalisation and resource-constrained deployment (Gu et al. 2024). The present thesis does not solve these deployment problems; instead, it treats them as limits on the scope of conclusions drawn from synthetic SUMO experiments.

2.4 Conventional Traffic Signal Control

2.4.1 Fixed-Time Control

Fixed-time control assigns phase sequences and green durations in advance. Its main advantages are simplicity, predictability and ease of coordination where traffic

patterns are sufficiently stable. In coordinated corridors, offsets can be selected so that platoons encounter successive green phases. The limitation is that a timing plan does not directly respond to unanticipated short-term changes in demand.

A fixed plan should not automatically be treated as a weak or unrealistic comparator. Classical traffic engineering includes systematic procedures for selecting cycle length and green allocation from demand estimates. Webster-style signal timing is one example of demand-informed fixed-time optimisation and remains influential as a benchmark concept (Webster 1958). Consequently, comparing RL only with an arbitrary static cycle can exaggerate the apparent benefit of learning.

This distinction is central to the final thesis experiment. The Original Fixed-Time controller is retained for continuity, but a Webster-derived Fixed-Time controller is added so that the learned policies are not judged only against the original 90-second programme.

2.4.2 Actuated Control

Actuated traffic control changes phase duration in response to detector calls. The Federal Highway Administration defines actuated signal control as a mode in which the time assigned to each phase is at least partly controlled by detector actuations (Federal Highway Administration 2008). In contrast to a fully pre-timed plan, an actuated phase can extend when traffic is present and terminate when demand disappears.

Actuated control is important for RL evaluation because it is adaptive without requiring a learned neural-network policy. If an RL controller cannot outperform a correctly configured actuated system under the same simulated traffic conditions, the practical case for introducing learning is weaker than a comparison with static timing alone would suggest.

The final experiment therefore treats Actuated control as the strongest conventional reference under the tested medium-demand conditions.

2.4.3 Adaptive and Commercial Control

Operational adaptive systems can use network-level detection, optimisation and prediction beyond simple local actuation. Cabrejas-Egea et al. (2021) highlighted that RL traffic-signal research often lacked comparison with commercial controllers and therefore compared RL agents with systems including MOVA, SurTrac and cyclic control. Their work is methodologically important because it reframes the central question from “can RL beat a static plan?” to “can RL provide value against a strong system that practitioners could realistically use?”

This thesis does not reproduce MOVA or SurTrac. Nevertheless, the same principle is applied by including both Webster and Actuated baselines and by avoiding a universal-superiority claim when the strongest baseline performs better.

2.5 Reinforcement Learning Foundations for Traffic Signal Control

2.5.1 Markov Decision Process Formulation

A standard RL problem is represented as a Markov Decision Process (MDP) containing states, actions, transition dynamics, rewards and a discount factor (Sutton & Barto 2018). At decision time t , an agent observes a state s_t , selects an action a_t , receives reward r_t , and transitions to the next state s_{t+1} . The objective is to maximise expected discounted return,

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k},$$

where $\gamma \in [0, 1]$ controls the relative importance of future rewards.

Traffic-signal control does not perfectly satisfy the fully observed MDP assumption from the viewpoint of a local intersection. A controller usually sees only local detector information and perhaps limited neighbour information, while future arrivals depend on upstream traffic and other controllers. Network-level traffic control is therefore more naturally treated as partially observable, especially in decentralised MARL.

2.5.2 Deep Q-Networks

DQN combines Q-learning with neural-network function approximation. Mnih et al. (2015) demonstrated that a deep network can approximate action values while replay memory and a target network stabilise learning. For discrete traffic-signal actions, DQN is attractive because the controller can estimate the long-term value of a small set of phase decisions.

The advantages relevant to this thesis include a direct discrete-action formulation and experience replay, which reuses previous transitions. Limitations include sensitivity to reward design, exploration schedule and non-stationarity when several independent agents learn simultaneously.

2.5.3 Proximal Policy Optimisation

PPO is an on-policy policy-gradient method that constrains policy updates through a clipped surrogate objective (Schulman et al. 2017). Instead of learning a Q-value for every discrete action and selecting actions through an ϵ -greedy mechanism, PPO directly optimises a parameterised policy.

PPO has become popular because it is comparatively simple to implement and usually stable across a range of control tasks. However, its on-policy nature means that old trajectories are not reused in the same manner as DQN replay data. The empirical question in this thesis is therefore not whether one algorithm is theoretically better, but how DQN and PPO behave under the same traffic-control interface and fixed training budget.

2.6 From RL to Multi-Agent Reinforcement Learning

2.6.1 Why Multiple Agents Are Needed

A centralised controller could concatenate observations from every intersection and select a joint action. As network size grows, however, the joint action space expands rapidly. Chu et al. (2020) explicitly identify this dimensionality problem as a reason centralised RL becomes infeasible for large-scale ATSC.

MARL distributes control across local agents. Each intersection chooses an action using its own observation and possibly information received from neighbours. This reduces the per-agent action dimension but introduces partial observability and non-stationarity because the behaviour of neighbouring agents changes during learning.

2.6.2 Independent Learning

Independent learning treats every other agent as part of the environment. It is operationally simple: each agent owns its policy, receives a local reward and is trained without a central critic. The number of agents can increase without changing the dimensionality of each agent’s policy network.

The disadvantage is weak coordination. A decision that clears a local queue may increase downstream pressure or block a route needed by another intersection. Independent agents also experience a changing environment during training because neighbouring policies are changing simultaneously.

The architecture used in this thesis is intentionally close to this simple end of the MARL spectrum. Limited neighbour information is included in the observation, but no explicit message-passing, shared critic or parameter-sharing mechanism is used. This makes the experiment useful as a transparent baseline for asking how much scalability can be achieved before coordination quality becomes a dominant limitation.

2.6.3 Explicit Cooperation and Communication

Cooperative MARL attempts to mitigate the limitations of independent learning by sharing state, action, reward or latent representations.

CoLight (Wei, Xu, Zhang, Zheng, Zang, Chen, Zhang, Zhu, Xu & Li 2019) uses graph attention to learn communication among traffic signals and was evaluated on large networks containing many signals. Its significance is not only performance but architecture: the model treats neighbour influence as something that can be learned rather than manually fixed.

Haddad et al. (2022) integrate neighbour state, action and reward information into the DQN learning process. Their results show how explicit cooperation can be embedded directly in the objective used by a local agent.

Song et al. (2024) move further toward centralised training with decentralised execution. Their counterfactual actor-critic uses decentralised actors, a centralised critic and a scheduler that exchanges information between agents. The approach reflects a common MARL principle: richer information can be used during training while execution remains distributed.

These studies define an important contrast with the thesis architecture. The present work does not claim state-of-the-art coordination. Instead, it measures the performance and computational behaviour of a simpler independent formulation and identifies where more sophisticated cooperation may become necessary.

2.7 State Representation, Action Design and Reward Engineering

2.7.1 State Representation

The observation supplied to an RL traffic controller determines what information the policy can use. Noaeen et al. (2022) found wide variation across network-level RL studies, with queue size, phase state, vehicle count, vehicle position and speed among frequently used state elements.

A large state can represent traffic more completely but increases model complexity and data requirements. A compact state can improve efficiency and interpretability but may hide important downstream conditions. This trade-off becomes more severe in MARL because local observations cannot contain the full network state without effectively recreating a centralised controller.

The six-element local observation used in this thesis deliberately keeps input dimensionality fixed as network size grows. That property supports architectural scalability, but the growing delay gap relative to Actuated control on larger grids suggests that compact local information may also constrain behavioural scalability.

2.7.2 Action Space

Traffic-signal RL studies use several action formulations: selection of the next phase, extension of the current phase, direct green-time allocation, phase duration adjustments or combinations of these decisions.

Action semantics must be defined precisely. If an action described as “hold” actually allows an underlying fixed programme to progress, then the learned policy does not control the signal in the way the mathematical action description implies. Such mismatches affect reproducibility and invalidate comparison between models trained under different implementations.

The final environment in this thesis therefore uses explicit HOLD and SWITCH semantics with a minimum-green constraint and deterministic yellow transition. This design prioritises transparent and auditable behaviour over a richer action space.

2.7.3 Reward Design

Reward construction is one of the most consequential choices in RL traffic-signal control. A reward based only on local queue may reduce queue at one approach while increasing network travel time elsewhere. A throughput-only reward may encourage aggressive switching or neglect fairness.

PressLight (Wei, Chen, Zheng, Wu, Gayah, Xu & Li 2019) is influential because it connects RL reward design to max-pressure traffic theory rather than choosing reward terms purely heuristically. The paper argues that a concise pressure-based representation can support a traffic-grounded optimisation objective.

Bouktif et al. (2023) similarly emphasise consistency between state and reward design and show that the definition of these elements materially affects training stability and final controller performance.

These studies motivate the interpretable composite reward used in the thesis. Waiting-time change, queue size, local discharge and switching cost are combined so that the components correspond to recognisable traffic objectives. The reward

remains a modelling choice rather than a theoretically optimal network objective, which is why final conclusions are based on external traffic metrics rather than cumulative reward.

2.8 Scalability in MARL Traffic Signal Control

2.8.1 Architectural Scalability

Architectural scalability concerns whether the controller can be instantiated on a larger network without redesigning network-specific model inputs or hard-coded agent identities.

Chu et al. (2020) present decentralised multi-agent A2C for both a large synthetic grid and a real-world Monaco network and explicitly frame decentralisation as a response to the dimensionality of centralised control. Their work provides a strong precedent for treating large-scale traffic control as a MARL problem rather than a single-agent extension.

CoLight (Wei, Xu, Zhang, Zheng, Zang, Chen, Zhang, Zhu, Xu & Li 2019) similarly demonstrates that graph-based communication can be structured so that a controller learns neighbour influence without relying on fixed neighbour indices.

The architecture in this thesis takes a simpler approach: intersections and neighbours are discovered from topology, and each agent retains the same local input and binary action definition. Successful instantiation from 4 to 25 agents therefore establishes architectural scalability within the tested synthetic networks.

2.8.2 Behavioural Scalability

Behavioural scalability asks whether controller quality remains acceptable as the network becomes larger. This is harder to measure because larger networks may also contain longer routes, different demand distributions and different congestion severity.

A study that simply reports successful execution at a larger grid has not demonstrated behavioural scalability. The relevant question is whether performance relative to a matched baseline remains stable and whether throughput, travel time, waiting time and queues degrade in a meaningful way.

Bie et al. (2024) add another layer to this issue by showing that intersection heterogeneity matters. Real urban networks contain different geometry, demand and signal timing characteristics, while many MARL experiments use homogeneous grids. A homogeneous 5×5 test therefore demonstrates only one dimension of scalability.

The final thesis reflects this distinction. Demand is calibrated separately for

each network size, and controller comparisons are interpreted primarily within each grid rather than treating absolute cross-grid queue or travel-time values as directly comparable.

2.8.3 Computational Scalability

Computational scalability concerns training time, memory, model storage and online decision latency.

Wu et al. (2022) combine distributed DRL with a distributed computing architecture, illustrating that large-scale traffic-signal control is not only an algorithmic problem but also a systems problem.

Independent agents have a predictable computational consequence: model count increases with intersection count. DQN additionally requires separate replay storage per agent, while PPO maintains rollout data. Whether this is practical depends on the slope of resource growth and the real-time decision budget.

This thesis therefore measures training runtime, peak RAM, total buffer memory, total model size and joint deterministic inference latency rather than inferring computational scalability from algorithm structure alone.

2.9 Coordination Versus Scalability

The literature reveals a recurring tension. Rich communication and centralised critics can improve coordination but increase information requirements and implementation complexity. Independent agents are easier to scale but observe less of the network.

The comparison between independent and cooperative learning is therefore not one-directional. Recent work by Ren et al. (2025) explicitly asks whether cooperative control is always better and investigates cases in which independent control can outperform cooperative approaches. This is an important corrective to the assumption that adding communication automatically improves a MARL system.

The thesis adopts an independent baseline precisely because it is interpretable and easy to scale. Its larger-grid results can then reveal whether the absence of stronger cooperation becomes visible in traffic efficiency.

2.10 Evaluation Metrics and the Problem of Reward-Only Reporting

Training reward is useful for diagnosing learning progress but is not a sufficient measure of traffic performance. A reward is constructed by the researcher and may

weight competing objectives in a way that conceals poor behaviour on an external metric.

For this reason, traffic-signal studies commonly report operational measures such as queue length, waiting time, delay, travel time, speed or throughput. Noaeen et al. (2022) show substantial variation in both state and reward elements across the literature, making direct comparison difficult unless common external metrics are retained.

The final thesis therefore reports five operational outcomes independently of the training objective: completion rate, average travel time, average waiting time, average time loss and mean sampled queue length. Teleports are treated as an integrity diagnostic rather than a performance reward.

This separation is methodologically important under severe oversaturation. If only completed trips are used for travel-time statistics, a controller with very low completion can appear deceptively efficient on the subset of trips that finished. Completion rate must therefore be interpreted together with delay measures.

2.11 Baseline Selection and Fair Comparison

The quality of the comparison baseline is one of the clearest weaknesses in RL traffic-signal evaluation.

Cabrejas-Egea et al. (2021) argue that the literature often lacks comparison with commercially available systems and compare their RL controllers with MOVA, SurTrac and cyclic control. This demonstrates that baseline strength is part of experimental validity, not a secondary presentation choice.

A weak baseline can produce a large percentage improvement even when the learned policy is not competitive with a reasonable adaptive controller. Conversely, a strong baseline can reveal that the principal contribution of MARL lies in flexibility, transferability or extensibility rather than immediate reduction in delay.

The revised thesis therefore uses a hierarchy of conventional references. Original Fixed-Time provides continuity with the initial experiment; Webster provides a stronger demand-informed fixed-time benchmark; Actuated provides a non-learning adaptive benchmark. The final interpretation is based primarily on the stronger comparisons.

2.12 Simulation Platforms, SUMO and the Sim-to-Real Gap

2.12.1 Why Microscopic Simulation Is Used

RL traffic control requires repeated interaction with traffic conditions, including poor exploratory actions that would be unacceptable in a live road network. Microscopic simulation therefore provides a controlled environment in which every vehicle can be modelled and the same scenarios can be repeated under different controllers.

SUMO is an open-source microscopic traffic simulator designed to model individual vehicles and large road networks (Alvarez Lopez et al. 2018). It is widely used in RL traffic-signal research because TraCI enables an external Python process to read simulator state and modify traffic-light behaviour during runtime.

The use of SUMO improves experimental repeatability but does not eliminate modelling assumptions. Vehicle behaviour, sensing and communication are idealised relative to a deployed traffic system.

2.12.2 Simulation Validity

A policy that performs well in simulation may exploit properties of the simulator that do not transfer to physical roads. Real systems contain detector noise, uncertain driver responses, incidents, communications delay, hardware constraints, pedestrian demands and safety rules.

The systematic review by Noaeen et al. (2022) notes the widespread use of non-commercial microscopic simulators and the limited interaction between RL research and transportation practitioners. This contributes to the broader sim-to-real gap.

The present research therefore restricts its conclusion to simulation-scale evidence and does not claim deployment readiness.

2.13 Reproducibility and Statistical Evaluation

2.13.1 Reproducibility

Reproducibility is particularly important in MARL because training and evaluation contain multiple sources of randomness: neural-network initialisation, exploration, sampled vehicle departures and stochastic traffic behaviour.

A reproducible study should report simulator configuration, network generation, route demand, algorithm hyperparameters, episode horizon, seeds, evaluation metrics and data-processing rules. Code and raw results further strengthen auditability.

Noaeen et al. (2022) explicitly include code availability and evaluation practice among the dimensions analysed in their systematic review, highlighting reproducibility as a field-level concern.

The final thesis addresses this by separating raw and derived results, retaining final model and evaluation artefacts, using fixed seed definitions and freezing authoritative numerical evidence by checksum before producing thesis tables and figures.

2.13.2 Repeated Evaluation and Statistical Testing

A single simulation run cannot establish whether a performance difference is robust to stochastic traffic variation. Repeated matched evaluation allows controller differences to be estimated under the same random seed realisations.

Statistical testing must also reflect the number of comparisons. If many algorithms, metrics and scenarios are tested independently at $\alpha = 0.05$, the probability of at least one false-positive conclusion rises. The revised thesis therefore applies Holm family-wise correction to predefined groups of paired tests rather than interpreting unadjusted p -values in isolation.

Small samples remain a limitation. Five paired seeds permit meaningful repeated comparison but provide limited inferential resolution. Consequently, the final analysis combines adjusted p -values with confidence intervals, effect size, median paired difference and exact sign-flip sensitivity.

2.14 Recent Directions: Heterogeneity, Interpretability and Integrated Control

2.14.1 Heterogeneous Networks

Bie et al. (2024) identify a limitation of homogeneous network assumptions and develop a MARL framework that explicitly incorporates intersection heterogeneity through spatiotemporal graph attention and value decomposition. Their work is directly relevant to the limits of the present thesis: a regular synthetic grid does not reproduce the diversity of geometry, demand and signal timing found in real networks.

This motivates future evaluation on irregular networks and supports cautious interpretation of the 5×5 result.

2.14.2 Interpretability

Interpretability has become increasingly important as RL traffic-control research moves closer to deployment.

Cai et al. (2025) propose XLight, an interpretable MARL framework that combines traffic-grounded reward design with regulatable policy functions. The motivation is practical as much as algorithmic: traffic agencies operate within safety, liability and regulatory constraints, so a high-performing opaque controller may remain difficult to deploy.

The present thesis does not solve the interpretability problem. Its contribution is narrower: the state variables, action semantics and reward components are explicitly documented and operational results are evaluated independently of training reward. The neural-network policies themselves remain opaque.

2.14.3 Integrated and Multimodal Control

Recent work also expands MARL beyond vehicle-only signal timing. Katzilieris et al. (2026) integrate traffic-signal control with dynamic bus-lane access using centralised training and decentralised execution. Such studies show that future adaptive traffic management may coordinate multiple infrastructure decisions rather than optimise signals in isolation.

This reinforces the need for scalable architectures but also increases the importance of multi-objective evaluation, fairness and safety—areas outside the present experiment.

2.15 Critical Synthesis

The reviewed literature supports five central conclusions.

First, RL traffic-signal performance is highly dependent on experimental design. State, action and reward definitions differ widely, and methods that are successful under one setup cannot automatically be assumed superior under another.

Second, decentralisation is a practical response to network scale, but it introduces partial observability and coordination problems. More complex cooperative methods attempt to recover network-level information through attention, communication, centralised critics or value decomposition.

Third, stronger coordination is not free. It increases implementation complexity, information requirements and sometimes computational cost, and recent work shows that cooperation is not necessarily superior in every setting.

Fourth, baseline strength is essential to valid interpretation. A large improvement

over a weak static plan is not equivalent to an improvement over demand-responsive or commercial traffic control.

Fifth, scalability must be separated into architectural, behavioural and computational dimensions. A controller can instantiate more agents successfully while losing traffic efficiency or becoming more expensive to train.

These conclusions directly shape the final experimental design of this thesis.

2.16 Research Gap

The literature contains highly sophisticated MARL controllers and large-scale traffic-signal experiments. The research gap addressed here is therefore not the absence of MARL traffic-signal control itself.

Instead, the gap lies in the combination of four evaluation requirements that are not consistently addressed together in a simple, reproducible independent-agent study:

1. **Baseline strength:** learned controllers should be compared not only with an arbitrary cyclic controller but also with demand-informed fixed-time and adaptive conventional control.
2. **Demand sensitivity:** controller conclusions should be tested across low, medium, changing and severe oversaturated demand rather than inferred from one favourable regime.
3. **Measured scalability:** architectural generalisation should be accompanied by behavioural and computational measurements as the number of controlled intersections increases.
4. **Reproducible inference:** repeated matched simulations, explicit multiplicity correction, effect-size reporting and traceable raw evidence should be used to distinguish robust differences from favourable individual runs.

The study therefore does not attempt to outperform every recent cooperative MARL method. Its contribution is to establish a transparent independent DQN/PPO benchmark and evaluate it under this stricter evidence framework.

2.17 Theoretical and Conceptual Framework

The theoretical framework combines decentralised sequential decision-making with traffic-performance evaluation.

At the agent level, each intersection is treated as an RL decision maker operating under partial observation. DQN represents value-based learning, while PPO represents policy-gradient learning. The agent receives a compact local observation, selects a HOLD or SWITCH action and receives a local composite reward.

At the network level, the collection of independent agents forms a MARL system whose joint behaviour emerges without a central critic. Limited neighbour information provides local context but not complete coordination.

At the evaluation level, controller quality is not defined by reward alone. Operational traffic outcomes provide the external criteria against which the learned policy is judged. Conventional controllers provide counterfactual reference systems, and repeated matched seeds provide the basis for statistical comparison.

At the scalability level, three questions are separated:

- can the same software architecture instantiate more agents?
- does relative traffic performance remain acceptable?
- do training, memory, storage and decision latency remain computationally practical?

This framework connects the literature directly to the experimental methodology in Chapter 3.

2.18 Chapter Summary

This chapter established the theoretical and empirical foundation for the final experiment. A structured literature search was documented using verified peer-reviewed and authoritative sources, while avoiding the unsupported claim that the thesis itself constitutes a new systematic review.

Conventional traffic engineering shows why both fixed-time and actuated controllers remain important baselines. RL provides adaptive sequential decision-making, while MARL distributes that decision process across multiple intersections at the cost of partial observability and coordination difficulty. The literature demonstrates increasingly sophisticated solutions based on max-pressure rewards, graph attention, neighbour communication, value decomposition, centralised critics and interpretable policy structures.

The critical synthesis identified four requirements for the present study: strong conventional baselines, multiple demand regimes, explicit separation of architectural/behavioural/computational scalability, and reproducible statistical evaluation. These requirements lead directly to the final methodology described in Chapter 3.

Chapter 3 Research Methodology

3.1 Introduction

This chapter describes the final implemented methodology used to generate the experimental evidence reported in this thesis. The study used Python, SUMO and TraCI to implement an independent multi-agent reinforcement learning (MARL) framework for adaptive traffic signal control. The primary experiment used a 2×2 network under low, medium, high and dynamic traffic demand. A dedicated scalability experiment reused the same environment, observation structure, action semantics, reward formulation and training budget on 3×3 , 4×4 and 5×5 synthetic grids under independently calibrated medium-demand conditions.

The final experimental design compares five controllers: the original fixed-time controller, a Webster-derived fixed-time controller, an actuated controller, independent Deep Q-Network (DQN) agents and independent Proximal Policy Optimisation (PPO) agents. Learned controllers were trained with a fixed budget of 50,000 decision steps using training seed 1 and were evaluated deterministically on five matched SUMO seeds (11–15). The final evidence base contains 175 unique controller–scenario–network–seed evaluation runs.

3.2 Research Philosophy

A positivist research philosophy was adopted because the study evaluates controller performance using observable and repeatable simulation outputs rather than subjective interpretation. Waiting time, travel time, time loss, queue length, trip completion and computational resource measurements were treated as measurable variables. The conclusions are therefore restricted to the implemented synthetic networks, route definitions, controller settings and evaluation protocol.

3.3 Research Approach

The research followed a deductive and experimental approach. Existing reinforcement-learning and adaptive traffic-control literature motivated the expectation that learning-based controllers could respond to changing traffic conditions more effectively than a static fixed-time programme. However, the final experiment was designed not to assume that MARL would outperform all conventional methods. Instead, DQN and PPO were evaluated against three conventional baselines of increasing strength: the original fixed-time plan, a Webster-derived fixed-time plan and an actuated controller.

The same performance definitions and matched evaluation seeds were used within each controller comparison. Training, evaluation and statistical analysis were separated so that final claims were based on deterministic evaluation results rather than training reward alone.

3.4 Research Design

The study used a quantitative microscopic-simulation experiment with two complementary components.

The first component was a demand-sensitivity experiment on the 2×2 network. Five controllers were evaluated under low, medium, high and dynamic demand. With five matched evaluation seeds per controller and scenario, this component contains:

$$4 \text{ scenarios} \times 5 \text{ controllers} \times 5 \text{ seeds} = 100 \text{ runs.}$$

The second component was a medium-demand scalability experiment. The same five controllers were evaluated on 2×2 , 3×3 , 4×4 and 5×5 networks, corresponding to 4, 9, 16 and 25 controlled intersections. The 2×2 medium runs are already included in the first component; the three larger networks contribute:

$$3 \text{ additional grids} \times 5 \text{ controllers} \times 5 \text{ seeds} = 75 \text{ additional runs.}$$

The final evaluation therefore contains 175 unique runs.

3.5 Simulation Environment

SUMO was used as the microscopic traffic simulator and TraCI provided the runtime interface between Python and SUMO (Alvarez Lopez et al. 2018, Eclipse SUMO 2026a). The simulator advances internally in one-second increments, while each

MARL agent makes a control decision every five simulated seconds. All final experiments used the same verified SUMO installation and the same route and network files retained in the project.

Each final run lasted 1,800 simulated seconds. Evaluation outputs include SUMO trip information and step-level controller logs. The evaluation pipeline also records per-second vehicle departures, arrivals and teleport events so that completion and simulation-integrity diagnostics can be verified independently of trip-level summaries.

3.6 Overall MARL Architecture

Each signalised intersection is represented by one independent reinforcement-learning agent. The implementation does not use parameter sharing, a centralised critic or centralised training. Each DQN agent owns its own neural network and replay buffer, while each PPO agent owns its own policy/value parameters and rollout buffer. All agents interact with the same SUMO environment.

Traffic-light intersections and direct neighbouring traffic-light nodes are discovered automatically from each SUMO network. Consequently, the controller code does not depend on hard-coded intersection identifiers and can instantiate 4, 9, 16 or 25 independent agents from the selected network topology.

At each five-second decision point, every agent receives its local observation, selects an action, and the environment applies all requested signal decisions. SUMO is then advanced while legal green and yellow timing constraints are enforced. The next observation and local reward are returned to the relevant learning algorithm.

Figure 3.1 summarises the final system architecture and the information flow between SUMO, TraCI, the traffic-signal environment, independent agents and the evidence layer.

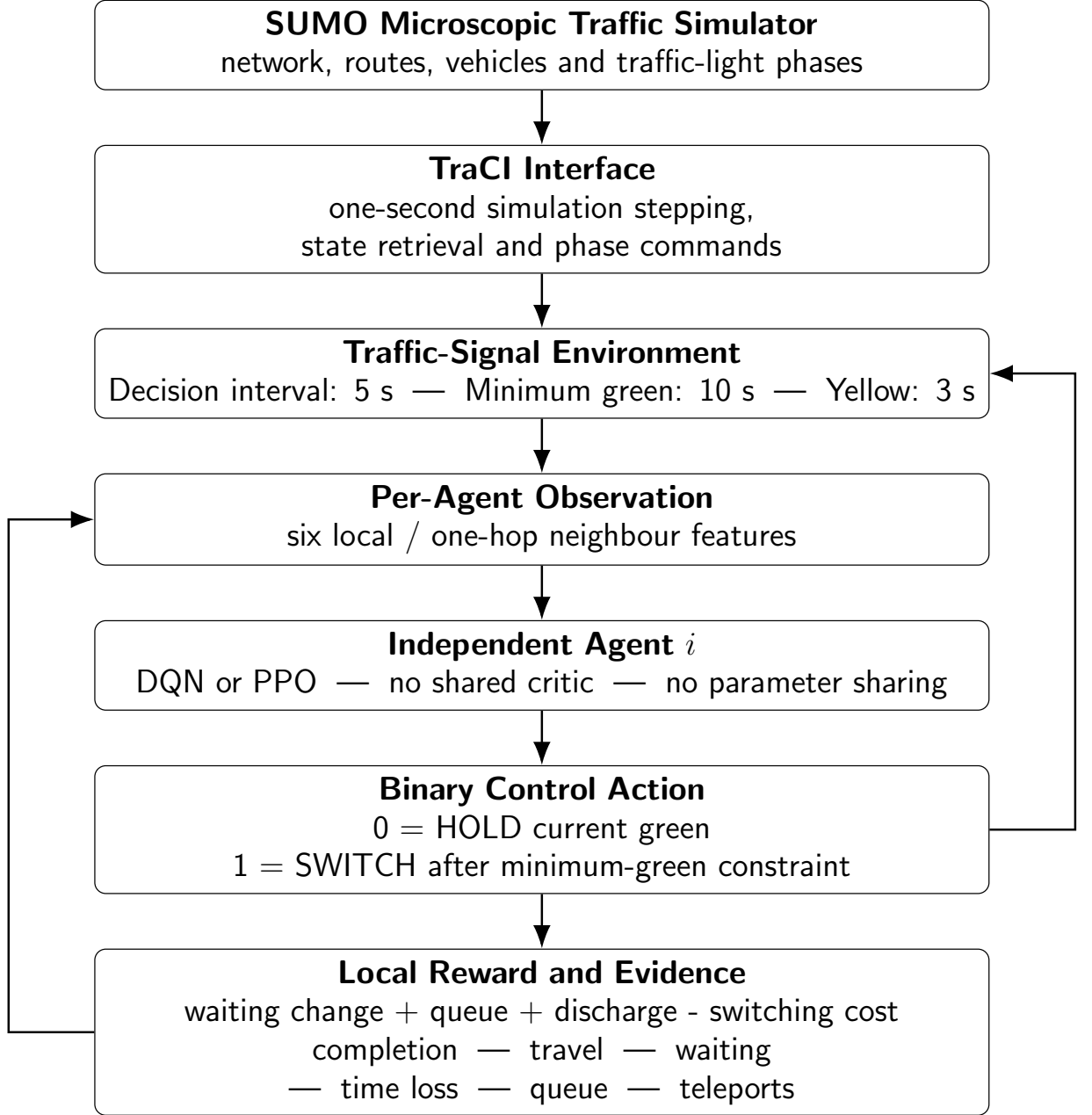


Figure 3.1: Final system architecture for independent MARL traffic-signal control.

Source: Author’s diagram based on the implemented Python–SUMO–TraCI architecture.

3.7 Traffic Networks and Demand Design

3.7.1 Primary 2×2 Demand Scenarios

The primary network contains four signalised intersections. Table 3.1 summarises the four route regimes retained for the final demand analysis and shows the planned demand, common episode duration and matched evaluation seeds.

The dynamic route consists of two temporal demand regimes within the same

Table 3.1: Primary 2×2 demand scenarios used in the final experiment.

Scenario	Planned vehicles	Episode	Evaluation seeds
Low	600	1,800 s	11–15
Medium	1,500	1,800 s	11–15
High	3,000	1,800 s	11–15
Dynamic	1,200	1,800 s	11–15

Source: Author’s final calibrated route and evaluation configuration.

1,800-second episode. It is interpreted as an in-distribution changing-demand test because the learned models were trained on the same route structure and were evaluated with different SUMO seeds. It is therefore not presented as out-of-distribution generalisation.

The high-demand scenario is intentionally severe and is treated as an oversaturated stress-test regime. Its purpose is to examine failure behaviour when demand approaches or exceeds the effective capacity of the synthetic network, rather than to represent typical medium congestion.

3.7.2 Scalability Demand Calibration

The scalability experiment uses medium-demand conditions on all four grid sizes, but planned vehicle counts are not scaled in direct proportion to the number of intersections. Network size changes total road length, route-length distribution, network storage and the number of alternative paths. A proportional increase in vehicles would therefore change both topology and congestion severity simultaneously, making controller-scale effects harder to interpret.

The final route files were calibrated independently. Planned vehicle count was an input to calibration, not the criterion used to declare two grids equally demanding. Table 3.2 reports the resulting route counts and experimental configuration.

Table 3.2: Final medium-demand scalability configuration.

Grid	Agents	Planned vehicles	RL training	Evaluation seeds
2×2	4	1,500	50,000 steps	11–15
3×3	9	1,800	50,000 steps	11–15
4×4	16	2,000	50,000 steps	11–15
5×5	25	1,566	50,000 steps	11–15

Source: Author’s final scalability configuration and calibrated route files.

To make the calibration criterion observable rather than asserted, Table 3.3 reports the five-seed Webster-derived fixed-time outcomes on the four final medium-demand route sets. The same controller is used as the diagnostic reference so that calibration is assessed independently of the learned policies.

Table 3.3: Fixed-time calibration check for the final medium-demand scalability routes.

Grid	Planned vehicles	Completion (%)	Waiting (s)	Time loss (s)
2×2	1,500	94.27	15.13	36.28
3×3	1,800	93.58	16.10	39.33
4×4	2,000	92.98	15.02	39.13
5×5	1,566	92.36	15.18	40.87

Source: Author’s five-seed Webster-derived fixed-time results from the final medium-demand scalability experiment.

The calibration check shows that average waiting time remains within 15.02–16.10 s across the four grids, while completion remains within 92.36–94.27% and time loss within 36.28–40.87 s. This supports the interpretation that the route sets generate a broadly comparable medium-demand service regime under a common fixed-time reference. In particular, the 5×5 network should not be interpreted as lighter solely because its planned vehicle count (1,566) is lower than the 4×4 count (2,000): absolute vehicle count is topology-dependent. Travel time is not used as the primary calibration check because larger networks naturally alter route-length distributions.

Because demand is calibrated separately for each network, absolute completed-trip counts are not interpreted directly across grid sizes. Cross-grid controller behaviour is instead reported using trip-completion percentage together with travel time, waiting time, time loss and mean sampled queue length.

3.8 State Representation

Every agent receives the same six-element observation vector regardless of total network size. The observation contains:

1. local halted-vehicle count;
2. local accumulated lane waiting time;
3. mean lane occupancy;
4. current traffic-signal phase encoded numerically;
5. elapsed time in the current green phase;
6. mean queue condition of directly connected traffic-light neighbours.

Queue, waiting-time and timing features are scaled before being passed to the neural networks. Because the observation dimension remains constant as the grid expands, increasing network size increases the number of independent agents but not the dimensionality of each agent’s local input.

Table 3.4 makes the fixed per-agent representation explicit. The same six inputs are used at every tested network size.

Table 3.4: Six-element per-agent observation representation used at every tested network size.

No.	Feature	Analytical role
1	Local halted-vehicle count	Represents the current local queue on controlled approaches.
2	Accumulated lane waiting time	Captures persistent local delay rather than vehicle count alone.
3	Mean lane occupancy	Represents utilisation of the controlled incoming lanes.
4	Encoded signal phase	Gives the policy the current traffic-light state.
5	Elapsed green time	Supports timing-aware HOLD/SWITCH decisions and minimum-green compliance.
6	Mean neighbour queue condition	Supplies limited one-hop network context without a centralised state vector.

Source: Author’s final traffic-signal-environment observation implementation.

3.9 Corrected Action Space and Signal Semantics

The final environment uses a validated two-action control space. Table 3.5 makes the operational meaning of each action and the signal-safety timing constraints explicit.

Table 3.5: Final binary action semantics and signal-safety constraints.

Action / rule	Final meaning	Constraint
Action 0	Hold the current green	Green remains active
Action 1	Request transition to opposite green	Minimum green required
Minimum green	10 s	Enforced before transition
Yellow interval	3 s	Exact transition duration
All-red	0 s	No dedicated all-red phase
Decision interval	5 s	Agent decision frequency

Source: Author’s validated traffic-signal-environment implementation and phase-control regression test.

Action 0 explicitly holds the current green by extending the active green phase. It does not allow the underlying static SUMO programme to advance automatically. Action 1 requests a transition only after at least 10 seconds of green have elapsed.

When permitted, the environment initiates the corresponding three-second yellow phase and then activates the opposite green. This behaviour was validated with a dedicated phase-control regression test before final training and evaluation.

This correction is methodologically important because models trained under the earlier automatic-programme semantics are not behaviourally equivalent to models trained under the final HOLD/SWITCH semantics. All reported learned-controller results were therefore generated from models retrained under the corrected environment.

3.10 Reward Function

Each agent receives a local reward combining waiting-time change, queue size, local discharge and switching cost:

$$r_{i,t} = -\alpha\Delta W_{i,t} - \beta Q_{i,t} + \delta T_{i,t} - \lambda C_{i,t},$$

where $\Delta W_{i,t}$ is the change in accumulated local waiting time, $Q_{i,t}$ is the local halted-vehicle count, $T_{i,t}$ is a local discharge measure, and $C_{i,t}$ indicates a permitted agent-requested transition.

The implemented weights are:

$$\alpha = 0.3, \quad \beta = 0.2, \quad \delta = 0.1, \quad \lambda = 0.5.$$

The local discharge term counts vehicle identifiers that were present on the controlled approach lanes at the previous decision point but are no longer present after the next control interval. This avoids treating vehicles that merely remain on an approach as discharged traffic.

3.11 Learning Algorithms

The two learning algorithms provide complementary value-based and policy-gradient formulations. Table 3.6 summarises the methodological differences relevant to the implemented controller.

Table 3.6: Comparison of DQN and PPO characteristics relevant to the implemented traffic-signal controller.

Characteristic	DQN	PPO
Learning family	Value-based, off-policy	Policy-gradient, on-policy
Policy output	Action values for discrete HOLD/SWITCH actions	Probability distribution over HOLD/SWITCH actions
Experience use	Replay buffer reuses stored transitions	Rollout buffer uses recent on-policy trajectories
Exploration	ϵ -greedy schedule during training	Stochastic policy with entropy regularisation
Stabilisation	Target network and replay buffer	Clipped policy objective and Generalised Advantage Estimation (GAE)
Final deterministic evaluation	Highest-valued action	Highest-probability action
Primary implementation trade-off	Larger replay-memory footprint	Lower data reuse and larger saved model per agent

Source: Author’s synthesis of the implemented DQN and PPO configurations, based on Mnih et al. (2015) and Schulman et al. (2017).

3.11.1 Independent DQN

One independent Stable-Baselines3 DQN model is instantiated per traffic-light agent (Raffin et al. 2021), following the DQN value-learning principle introduced by Mnih et al. (2015). The final DQN configuration uses a learning rate of 10^{-3} , discount factor $\gamma = 0.95$, replay-buffer capacity of 50,000 transitions, batch size 64, and learning starts after 500 decision steps. A gradient update is performed every four decision steps. The exploration rate decays linearly from 1.0 to 0.05 over the first 70% of training. The target network is updated every 250 decision steps.

3.11.2 Independent PPO

One independent PPO model is instantiated per traffic-light agent, using the clipped policy-update formulation of Schulman et al. (2017). PPO uses a learning rate of 3×10^{-4} , discount factor $\gamma = 0.95$, rollout length of 128 steps, batch size 32, four optimisation epochs per rollout, Generalised Advantage Estimation (GAE) parameter $\lambda = 0.9$, clipping range 0.2, and entropy coefficient 0.01.

The final hyperparameters were held fixed across grid sizes for each algorithm. Table 3.7 provides a side-by-side comparison.

Table 3.7: Final DQN and PPO hyperparameter configuration used in the training runs.

Hyperparameter	DQN	PPO
Learning rate	1×10^{-3}	3×10^{-4}
Discount factor γ	0.95	0.95
Batch size	64	32
Replay-buffer capacity	50,000 transitions	Not applicable
Learning starts	500 decision steps	Not applicable
Gradient / rollout schedule	Gradient update every 4 decision steps	128-step rollout
Exploration / entropy	ϵ : 1.0 $\rightarrow$ 0.05 over first 70%	Entropy coefficient 0.01
Target update interval	250 decision steps	Not applicable
GAE λ	Not applicable	0.9
Clip range	Not applicable	0.2
Optimisation epochs	Not applicable	4 per rollout
Training budget	50,000 decision steps	50,000 decision steps
Training seed	1	1

Source: Author’s final DQN and PPO training configurations.

3.11.3 Training Protocol

Each algorithm–grid–scenario training job uses training seed 1 and a fixed budget of 50,000 decision steps. The final model at the end of this budget is retained; there is no post-hoc selection of a more favourable checkpoint from the evaluation seeds.

The primary 2×2 experiment requires four DQN and four PPO training jobs. The larger-grid medium-demand experiment requires three additional DQN and three additional PPO jobs. The final study therefore contains 14 learned-controller training jobs.

Figure 3.2 shows the complete final workflow from scenario construction and structural validation to learning, evaluation, evidence freezing and statistical analysis.

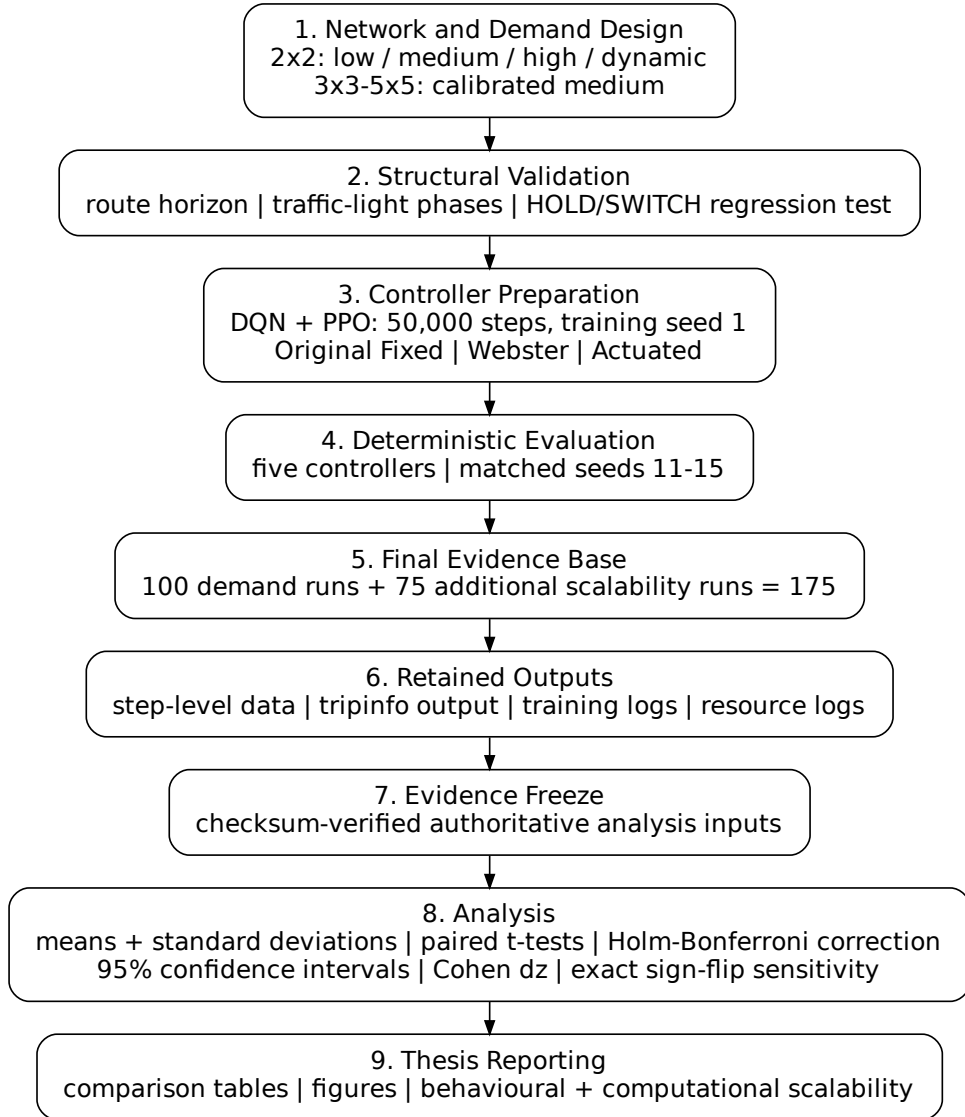


Figure 3.2: Final experimental workflow from validated scenario generation to frozen evidence and statistical analysis.

Source: Author's diagram based on the final experimental pipeline.

3.12 Conventional Control Baselines

Three conventional controllers are retained so that learned-controller performance is not judged only against a weak static reference.

3.12.1 Original Fixed-Time Controller

The original fixed-time programme uses two principal 42-second green phases separated by three-second yellow transitions, resulting in a 90-second cycle. This controller is retained because it represents the original baseline used during development and provides continuity with the initial experiment.

3.12.2 Webster-Derived Fixed-Time Controller

A stronger fixed-time baseline was generated using SUMO’s traffic-light cycle-adaptation tool, which modifies cycle length and green allocation according to Webster-style timing principles (Webster 1958, Eclipse SUMO 2026*b*). The final generated programmes were standardised to three-second yellow transitions, zero dedicated all-red time, four seconds of lost time, a minimum green of 10 seconds, and cycle-length bounds of 20–120 seconds. The final cycle length varies with route demand and network configuration rather than remaining fixed at 90 seconds.

3.12.3 Actuated Controller

An actuated baseline was generated using SUMO’s documented actuated traffic-light logic, representing detector-responsive signal operation consistent with standard actuated-control principles (Federal Highway Administration 2008, Eclipse SUMO 2026*c*). The final controller uses a minimum green of 10 seconds, a maximum green of 50 seconds, three-second yellow transitions and no dedicated all-red phase. This baseline responds to detected traffic conditions but does not use reinforcement learning.

The Webster and actuated controllers provide substantially stronger conventional references than the original fixed-time plan and are therefore central to the final interpretation of MARL performance.

3.13 Evaluation Metrics

Five operational metrics are used in the final inferential analysis:

1. trip completion rate (%);
2. average travel time (s);
3. average waiting time (s);
4. average time loss (s);

5. mean sampled queue length.

Trip completion is calculated from the number of trips completed within the 1,800-second evaluation horizon relative to planned demand. Average travel time, waiting time and time loss are calculated from completed SUMO tripinfo records. Mean queue length is calculated from controlled-lane halted-vehicle counts sampled during evaluation. Teleport counts are retained as a simulation-integrity diagnostic rather than as a primary performance outcome.

Training reward is reported only as a learning diagnostic and is not used as the final measure of traffic performance.

Table 3.8 summarises the direction and interpretation of the final evaluation measures.

Table 3.8: Operational and diagnostic performance metrics used in the final study.

Metric	Desired direction	Role in final evaluation
Trip completion rate	Maximise	Percentage of planned trips completed within the 1,800-s horizon.
Average travel time	Minimise	End-to-end time for trips completed within the horizon.
Average waiting time	Minimise	Trip-level time spent waiting according to SUMO tripinfo.
Average time loss	Minimise	Delay relative to ideal travel, including traffic-control effects.
Mean sampled queue length	Minimise	Halted vehicles sampled on controlled approaches during evaluation.
Teleport count	Zero preferred	Simulation-integrity diagnostic; not a primary performance objective.
Training cumulative reward	Diagnostic only	Used to inspect learning behaviour, not to rank final controllers.

Source: Author’s final evaluation and analysis definitions.

3.14 Statistical Analysis

Final controller comparisons use paired evaluation seeds 11–15. Pairing is appropriate because controllers within the same network and demand condition are evaluated on matched stochastic SUMO realisations.

For each scenario–metric family in the 2×2 demand experiment, seven predefined two-sided paired comparisons are tested:

1. DQN versus original fixed-time;

2. DQN versus Webster;
3. DQN versus actuated;
4. PPO versus original fixed-time;
5. PPO versus Webster;
6. PPO versus actuated;
7. DQN versus PPO.

The same seven comparisons are used within each grid-metric family in the medium-demand scalability analysis. Five metrics and four scenarios produce 140 demand-analysis comparisons; five metrics and four grid sizes produce another 140 scalability comparisons.

The paired t -test is the primary parametric test. Family-wise multiplicity is controlled using the Holm–Bonferroni step-down correction (Holm 1979) separately within each seven-comparison scenario-metric or grid-metric family. The analysis also reports the paired mean difference, 95% confidence interval, median paired difference and an exact two-sided paired sign-flip sensitivity test.

For magnitude reporting, the paired-samples standardised effect size Cohen’s d_z is used. For paired differences d_i , it is defined as

$$d_z = \frac{\bar{d}}{s_d},$$

where $\bar{d}$ is the mean paired difference and s_d is the sample standard deviation of the paired differences. The subscript z makes the difference-score standardiser explicit for a within-pair design (Cohen 1988, Lakens 2013).

Only five seed pairs are available per comparison. With $2^5 = 32$ possible sign assignments, the smallest attainable exact two-sided sign-flip probability is 0.0625. For this reason, the exact sign-flip analysis is treated as a small-sample sensitivity check rather than as a replacement for the paired t -test. Inferential results are interpreted together with effect magnitude, confidence intervals and cross-seed consistency.

3.15 Computational Scalability Measurement

Traffic performance alone does not establish whether the independent-agent architecture remains computationally practical as the number of intersections increases. The final scalability analysis therefore records computational measurements for DQN and PPO at 4, 9, 16 and 25 agents.

The retained measurements include:

- total training runtime;
- peak process-tree Random Access Memory (RAM) during training;
- total replay-buffer or rollout-buffer memory;
- total saved model size;
- mean deterministic joint decision latency;
- 95th-percentile joint decision latency;
- peak evaluation process-tree RAM.

Central Processing Unit (CPU) and Graphics Processing Unit (GPU) utilisation percentages were not instrumented during the frozen final runs. They are therefore not reconstructed retrospectively. The computational analysis reports measurements that were actually captured—wall-clock training/evaluation runtime, RAM, buffer memory, model size and inference latency—and treats missing processor-utilisation percentages as a limitation.

Joint inference latency is additionally expressed as a percentage of the 5,000-ms decision interval. Simple least-squares fits against agent count are used only as descriptive summaries of the four observed network sizes. They are not interpreted as asymptotic computational-complexity proofs.

3.16 Data Collection and Reproducibility

Raw experimental outputs are preserved separately from derived analysis files. The retained evidence includes route files, network files, final model files, training logs, step-level evaluation data files, SUMO tripinfo output files, conventional-controller summaries, learned-controller summaries, statistical-analysis outputs and computational-resource summaries.

The final numerical evidence was frozen after completion of the full analysis pipeline using Secure Hash Algorithm 256-bit (SHA-256) checksums. Thesis-ready tables and figures are generated only from checksum-verified source data files. This prevents later manual transcription from silently changing reported values.

Matched evaluation seeds, fixed episode duration, validated route horizons, validated traffic-light phase structure and explicit controller configuration support reproducibility within the submitted synthetic environment.

3.17 Threats to Validity

3.17.1 Internal Validity

Internal validity is strengthened by matched seeds, fixed route definitions within each comparison, identical evaluation horizons, deterministic learned-controller evaluation and frozen final configurations. Remaining threats include neural-network initialisation, hyperparameter sensitivity, reward-weight choice and independent-agent non-stationarity.

3.17.2 Construct Validity

Traffic performance is represented using several complementary operational metrics rather than training reward alone. This reduces the risk that an apparent improvement in the learning objective is mistaken for improved traffic operation. However, emissions, pedestrian delay, fairness, public-transport priority and safety are outside the final objective.

3.17.3 External Validity

All networks are synthetic SUMO grids. The largest tested network contains 25 signalised intersections and the larger-grid experiment is limited to independently calibrated medium demand. The results therefore support claims only within the tested synthetic range and do not establish city-scale deployment or real-world field performance.

3.17.4 Statistical Validity

Five matched evaluation seeds provide repeated observations but remain a small sample. Confidence intervals, effect sizes and paired tests can therefore be sensitive to small run-to-run differences. Holm–Bonferroni correction reduces the risk of inflated family-wise false positives, but the study does not claim that $n = 5$ provides definitive population-level inference.

3.17.5 Simulation Assumptions

SUMO provides idealised state access compared with field detector systems. Real deployments would introduce detector error, communication delays, heterogeneous driver behaviour, incidents, pedestrians, public transport and actuator faults. No final result is interpreted as evidence of deployment readiness.

3.18 Chapter Summary

This chapter defined the final methodology used for the revised thesis. The experimental design compares three conventional controllers and two independent MARL controllers across four 2×2 demand conditions and four medium-demand grid sizes. The corrected environment uses explicit HOLD/SWITCH signal semantics, a 10-second minimum green, a three-second yellow transition and a fixed six-element local observation. Fourteen learned-controller training jobs and 175 deterministic evaluation runs form the final empirical evidence base. Controller performance is assessed using operational traffic metrics, paired seed comparisons, Holm-adjusted statistical testing and computational-scalability measurements. The next chapter reports the resulting demand-sensitivity, scalability and computational findings.

Chapter 4 Results

4.1 Introduction

This chapter reports the final results produced by the corrected final experimental pipeline described in Chapter 3. The evidence base contains 175 unique deterministic evaluation runs. The 2×2 demand study contributes 100 runs (four demand scenarios $\times$ five controllers $\times$ five matched seeds), and the 3×3 , 4×4 and 5×5 medium-demand scalability extension contributes 75 additional runs. Five controllers are compared throughout: Original Fixed-Time, Webster Fixed-Time, Actuated, DQN and PPO.

Operational conclusions are based on trip completion rate, average travel time, average waiting time, average time loss and mean sampled queue length. Training reward is treated only as a learning diagnostic. All inferential comparisons use matched SUMO seeds 11–15 and the Holm-adjusted paired-test procedure defined in Chapter 3. The chapter first examines demand sensitivity on the primary 2×2 network, then evaluates traffic and computational scalability from 4 to 25 agents.

4.2 Final Evaluation Overview

The revised evaluation differs materially from the initially submitted three-controller analysis because two stronger conventional baselines were added and all learned models were retrained under the corrected HOLD/SWITCH signal semantics. The resulting comparison therefore tests whether the learned controllers outperform not only the original fixed-time programme but also a Webster-derived fixed-time plan and an adaptive actuated controller.

No teleport events were recorded in the final summary data for any controller in the reported demand or medium-demand scalability experiments. This supports the internal consistency of the simulation runs, although absence of teleports does not by itself establish real-world validity.

4.3 Demand Sensitivity on the 2×2 Network

Table 4.1 reports five-seed mean performance for every controller–scenario combination. The table is presented before scenario-specific interpretation so that controller ranking can be assessed using the same metrics across all four demand regimes.

Table 4.1: Final 2×2 demand-sensitivity results. Values are five-seed means.

Scenario	Controller	Completion (%)	Travel (s)	Waiting (s)	Time loss (s)	Queue
Low	Original Fixed	94.93	97.43	19.57	33.80	1.627
Low	Webster	95.03	83.73	6.32	20.05	0.468
Low	Actuated	95.37	81.59	4.81	17.88	0.356
Low	DQN	95.17	83.15	6.11	19.43	0.383
Low	PPO	95.07	83.07	5.97	19.34	0.377
Medium	Original Fixed	92.95	122.23	36.24	59.56	6.513
Medium	Webster	94.27	98.89	15.13	36.28	2.315
Medium	Actuated	94.43	96.62	14.85	33.95	2.128
Medium	DQN	94.13	98.48	15.49	35.85	2.192
Medium	PPO	94.12	100.96	17.07	38.34	2.511
High	Original Fixed	13.44	273.54	162.00	200.89	74.092
High	Webster	15.53	258.63	134.48	187.96	70.123
High	Actuated	13.17	288.17	177.36	215.33	73.682
High	DQN	14.55	261.99	139.49	189.10	72.385
High	PPO	7.45	358.44	233.98	259.01	74.152
Dynamic	Original Fixed	93.02	116.10	28.95	50.18	4.334
Dynamic	Webster	93.95	102.84	14.54	36.87	2.005
Dynamic	Actuated	94.40	94.47	10.69	28.49	1.361
Dynamic	DQN	93.97	97.19	12.36	31.22	1.597
Dynamic	PPO	93.28	104.20	16.09	38.28	2.149

Source: Author’s analysis of the frozen final simulation results.

4.3.1 Low Demand

Under low demand, all controllers completed approximately 95% of planned trips. Actuated control produced the best descriptive traffic-efficiency results, with 81.59 s average travel time, 4.81 s average waiting time and a mean queue of 0.356. DQN and PPO remained close to Webster and far better than the Original Fixed-Time controller on delay-based metrics.

Relative to Original Fixed-Time, DQN reduced average travel time by 14.28 s and PPO by 14.36 s; both differences remained Holm-significant ($p_H < 0.001$). Waiting time was also lower by 13.46 s for DQN and 13.60 s for PPO ($p_H < 0.001$ in both cases). Against the stronger Actuated controller, however, DQN and PPO were significantly slower by 1.56 s and 1.48 s respectively ($p_H = 0.0126$), and both had significantly higher waiting time ($p_H = 0.0098$). Their small queue differences relative to Actuated were not Holm-significant. Thus, low demand demonstrates that the apparent MARL advantage depends strongly on baseline strength. Figure 4.1 visualises the travel-time separation among all five controllers across the four demand regimes and makes the high-demand deterioration particularly visible.

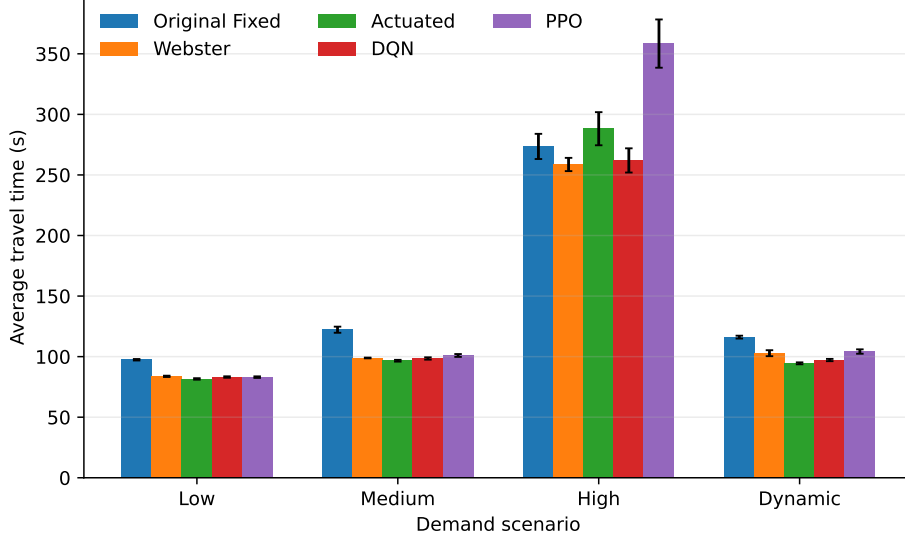


Figure 4.1: Average travel time across demand scenarios. Error bars show standard deviation across five matched evaluation seeds.

Source: Author’s analysis of the frozen final simulation results.

4.3.2 Medium Demand

Medium demand produced the clearest comparison between DQN and the strongest conventional baseline. Actuated control again had the best descriptive means: 94.43% completion, 96.62 s travel time, 14.85 s waiting time and mean queue 2.128. DQN followed closely at 94.13% completion, 98.48 s travel time, 15.49 s waiting time and mean queue 2.192. Webster was also competitive, while PPO was weaker at 100.96 s travel time, 17.07 s waiting time and mean queue 2.511.

DQN versus Actuated was not Holm-significant for completion ($p_H = 0.6508$), travel time ($p_H = 0.1255$), waiting time ($p_H = 0.5984$), time loss ($p_H = 0.1284$) or queue length ($p_H = 0.3559$). PPO, in contrast, was significantly worse than Actuated for travel time (+4.34 s, $p_H = 0.0071$), waiting time (+2.22 s, $p_H = 0.0061$), time loss (+4.39 s, $p_H = 0.0068$) and queue length (+0.383, $p_H = 0.0055$), although completion rate was not significantly different. These results position DQN as competitive with the strong conventional controllers under medium demand, but do not support a claim of superiority. Figure 4.2 shows the corresponding waiting-time pattern and highlights the large PPO penalty under the high-demand stress test.

4.3.3 High-Demand Stress Test

The high-demand scenario produced qualitatively different behaviour and should be interpreted as a severe oversaturated stress test. Completion rates collapsed for all controllers; even the best descriptive result, Webster, completed only 15.53% of planned trips. Webster also recorded the lowest mean travel time (258.63 s), waiting

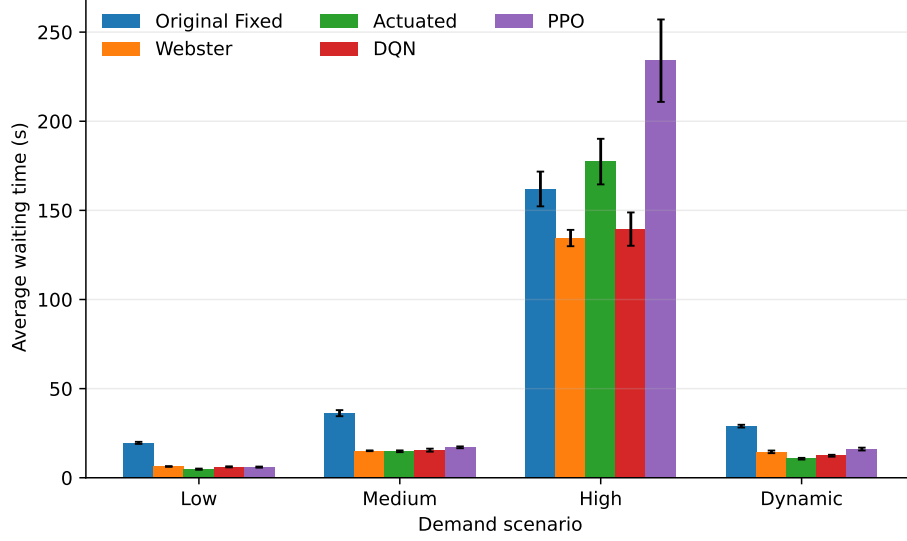


Figure 4.2: Average waiting time across demand scenarios. Error bars show standard deviation across five matched evaluation seeds.

Source: Author’s analysis of the frozen final simulation results.

time (134.48 s) and queue length (70.12) among the five controllers.

DQN remained comparatively robust. Its completion rate was 14.55%, with 261.99 s average travel time and 139.49 s waiting time. DQN did not differ significantly from Webster on completion, travel time or waiting time. It was, however, significantly better than Original Fixed-Time for travel time (-11.55 s, $p_H = 0.0244$) and waiting time (-22.51 s, $p_H = 0.0054$), and significantly better than Actuated on those same metrics.

PPO failed markedly under this regime. Its completion rate fell to 7.45%, while average travel time increased to 358.44 s and waiting time to 233.98 s. PPO completion was significantly lower than Original Fixed-Time (-5.99 percentage points, $p_H = 0.0024$), Webster (-8.08 , $p_H = 0.0002$) and Actuated (-5.73 , $p_H = 0.0016$). DQN also significantly outperformed PPO on completion ($+7.10$ percentage points, $p_H = 0.0024$), travel time (-96.46 s, $p_H = 0.0052$) and waiting time (-94.48 s, $p_H = 0.0051$). These results show that algorithm ranking changes under severe saturation and that PPO is particularly vulnerable in the tested high-demand regime.

4.3.4 Dynamic Demand

The dynamic experiment represents adaptation within a trained two-regime route structure rather than out-of-distribution generalisation. Actuated control again achieved the best descriptive values: 94.40% completion, 94.47 s travel time, 10.69 s waiting time and mean queue 1.361. DQN was the strongest learned controller at

93.97% completion, 97.19 s travel time, 12.36 s waiting time and mean queue 1.597.

DQN significantly outperformed Original Fixed-Time and Webster on travel time, waiting time and queue length, but remained significantly worse than Actuated on those same efficiency metrics. PPO was weaker than DQN: the DQN–PPO differences were significant for completion (+0.683 percentage points, $p_H = 0.0047$), travel time (−7.01 s, $p_H = 0.0008$), waiting time (−3.74 s, $p_H = 0.0008$) and queue length (−0.552, $p_H = 0.0016$). The dynamic results therefore support adaptation by the learned controllers relative to the weak static baseline, while also demonstrating that a strong actuated controller can remain superior.

4.4 Demand-Study Statistical Summary

Across the four demand scenarios, five metrics and seven predefined controller comparisons, 81 of 140 Holm-adjusted paired comparisons were significant at $\alpha = 0.05$. This count should not be interpreted as a score of overall controller quality; it summarises the amount of statistically detectable separation among controllers across a deliberately broad family of comparisons.

The paired t -test is the primary inferential procedure, while the exact two-sided sign-flip result is retained as a small-sample sensitivity analysis. With only five seed pairs, the exact two-sided sign-flip test has a minimum attainable p -value of 0.0625. Accordingly, the interpretation emphasises effect direction, confidence intervals, Holm-adjusted p -values and consistency across matched seeds rather than binary significance alone.

Table 4.2 condenses the principal learned-versus-Actuated inferential comparisons. Each cell reports the paired mean difference (learned controller minus Actuated) followed by the Holm-adjusted p -value in parentheses.

Table 4.2: Holm-adjusted learned-controller comparisons with Actuated control across the four 2×2 demand scenarios.

Scenario	Comparison	Completion Δ pp	Travel Δ s	Waiting Δ s	Time-loss Δ s	Queue Δ
Low	DQN–Actuated	−0.200 (1.0000)	+1.56 (0.0126*)	+1.30 (0.0098*)	+1.55 (0.0068*)	+0.027 (0.1314)
Low	PPO–Actuated	−0.300 (0.8217)	+1.48 (0.0126*)	+1.16 (0.0098*)	+1.47 (0.0081*)	+0.021 (0.1396)
Medium	DQN–Actuated	−0.293 (0.6508)	+1.86 (0.1255)	+0.64 (0.5984)	+1.90 (0.1284)	+0.064 (0.3559)
Medium	PPO–Actuated	−0.307 (0.6519)	+4.34 (0.0071*)	+2.22 (0.0061*)	+4.39 (0.0068*)	+0.383 (0.0055*)
High stress test	DQN–Actuated	+1.373 (0.1965)	−26.18 (0.0244*)	−37.87 (0.0054*)	−26.23 (0.0348*)	−1.297 (0.9627)
High stress test	PPO–Actuated	−5.727 (0.0016*)	+70.28 (0.0185*)	+56.62 (0.0163*)	+43.68 (0.0683)	+0.470 (1.0000)
Dynamic	DQN–Actuated	−0.433 (0.1703)	+2.72 (0.0109*)	+1.67 (0.0085*)	+2.73 (0.0120*)	+0.236 (0.0226*)
Dynamic	PPO–Actuated	−1.117 (0.0127*)	+9.73 (0.0008*)	+5.40 (0.0005*)	+9.80 (0.0008*)	+0.788 (0.0013*)

Notes: Difference = learned controller minus Actuated. Positive completion differences favour the learned controller; negative travel, waiting, time-loss and queue differences favour the learned controller. Parentheses contain Holm-adjusted paired- t p -values; * denotes $p_H < 0.05$. Source:

Author’s analysis of the frozen final paired-test outputs.

4.5 Medium-Demand Scalability from 4 to 25 Agents

The scalability analysis evaluates the same five controllers on networks containing 4, 9, 16 and 25 controlled intersections. Figure 4.3 shows how completion rate changes with agent count, while Figure 4.4 shows the corresponding travel-time trend. Because demand was calibrated independently for each grid, raw queue magnitude and absolute trip counts should not be interpreted as direct measures of whether larger grids are intrinsically easier or harder. The valid cross-grid interpretation is whether each controller retains its relative operational position within the calibrated network and whether computational cost remains manageable.

Table 4.3 reports the final five-seed means.

Table 4.3: Medium-demand scalability results across 4, 9, 16 and 25 agents. Values are five-seed means.

Grid	Agents	Controller	Completion (%)	Travel (s)	Waiting (s)	Time loss (s)	Queue
2x2	4	Original Fixed	92.95	122.23	36.24	59.56	6.513
2x2	4	Webster	94.27	98.89	15.13	36.28	2.315
2x2	4	Actuated	94.43	96.62	14.85	33.95	2.128
2x2	4	DQN	94.13	98.48	15.49	35.85	2.192
2x2	4	PPO	94.12	100.96	17.07	38.34	2.511
3x3	9	Original Fixed	91.69	135.94	40.06	63.84	4.086
3x3	9	Webster	93.58	111.52	16.10	39.33	1.507
3x3	9	Actuated	93.73	107.20	14.28	35.01	1.324
3x3	9	DQN	93.49	110.99	16.48	38.80	1.444
3x3	9	PPO	93.28	115.63	19.68	43.45	1.753
4x4	16	Original Fixed	91.57	144.55	40.10	62.54	2.650
4x4	16	Webster	92.98	121.26	15.02	39.13	0.909
4x4	16	Actuated	93.12	116.34	12.85	34.22	0.767
4x4	16	DQN	92.95	120.53	15.73	38.41	0.862
4x4	16	PPO	92.47	124.13	18.28	42.05	1.033
5x5	25	Original Fixed	91.26	156.21	43.14	64.89	1.484
5x5	25	Webster	92.36	132.23	15.18	40.87	0.484
5x5	25	Actuated	92.81	125.43	12.07	34.00	0.385
5x5	25	DQN	92.57	129.50	15.20	38.10	0.417
5x5	25	PPO	92.48	131.49	16.64	40.11	0.469

Source: Author’s analysis of the frozen final simulation results.

Actuated control achieved the best descriptive completion rate, travel time, waiting time and queue length at every tested grid size. DQN remained close to Webster and consistently stronger than PPO on the main delay metrics as network size increased. DQN travel time rose from 98.48 s at four agents to 129.50 s at 25 agents; PPO rose from 100.96 s to 131.49 s. These increases cannot be attributed purely to control degradation because larger networks also contain longer routes and independently calibrated demand.

4.5.1 Learned Controllers versus Actuated Control

At four agents, DQN was statistically indistinguishable from Actuated on all five analysed metrics after Holm correction. This changed as the network grew. At 9, 16

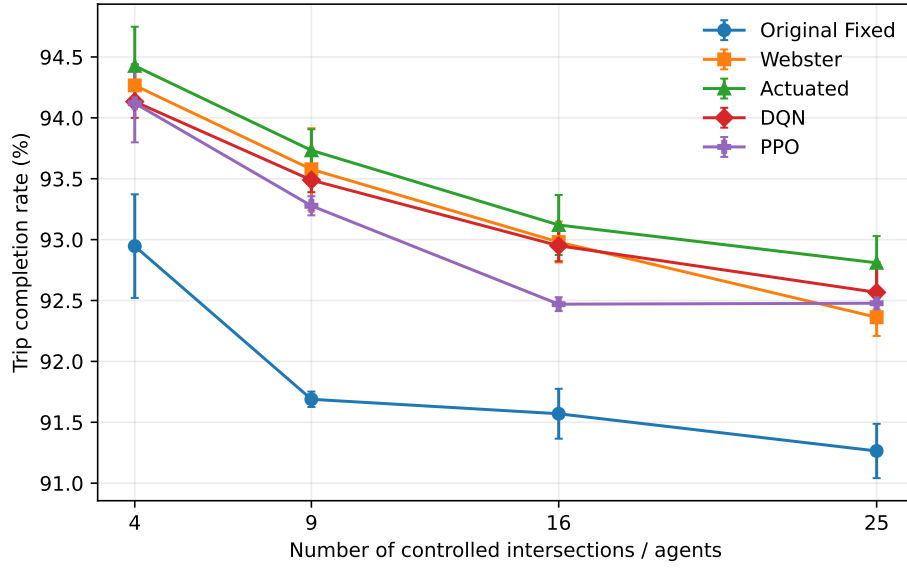


Figure 4.3: Trip completion rate across network sizes under calibrated medium demand. Error bars show standard deviation across five matched seeds.

Source: Author's analysis of the frozen final simulation results.

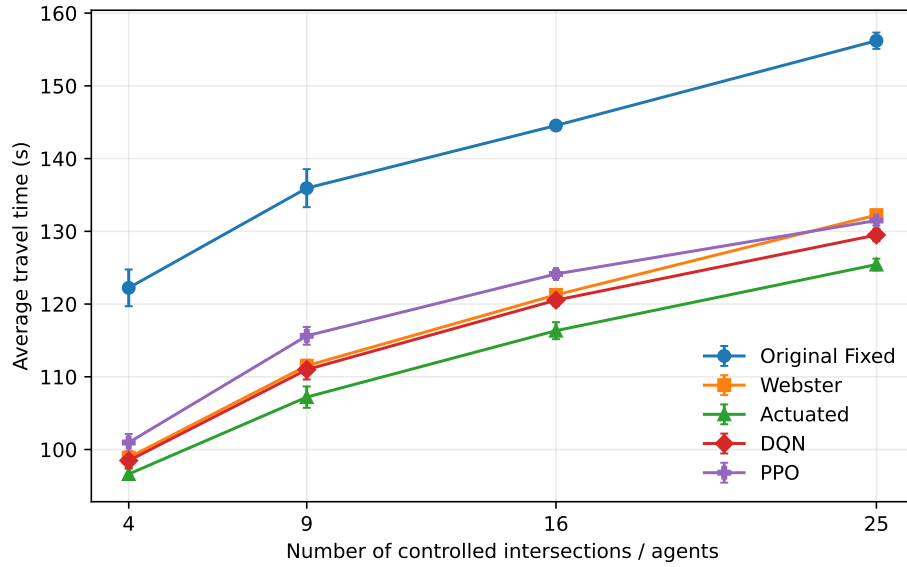


Figure 4.4: Average travel time across network sizes under calibrated medium demand. Error bars show standard deviation across five matched seeds.

Source: Author's analysis of the frozen final simulation results.

and 25 agents, DQN completion rate remained statistically indistinguishable from Actuated ($p_H = 0.2316, 0.3807$ and 0.4244 respectively), but DQN had significantly higher travel time, waiting time and queue length. At 25 agents, for example, DQN was 4.06 s slower ($p_H = 0.0054$), had 3.12 s more waiting time ($p_H = 0.0018$), and a queue difference of +0.033 ($p_H = 0.0478$). The queue result is statistically borderline and small in magnitude, so it is not treated as evidence of a practically large degradation.

PPO displayed a less favourable scaling pattern. At four agents, PPO was already significantly worse than Actuated on travel time, waiting time, time loss and queue length. At 9 and 16 agents its completion rate was also significantly lower. At 25 agents the completion-rate difference was not significant, but PPO still had significantly higher travel time (+6.05 s, $p_H = 0.0026$), waiting time (+4.57 s, $p_H = 0.0011$) and queue length (+0.084, $p_H = 0.0064$).

The most defensible behavioural-scalability conclusion is therefore that DQN preserves throughput comparatively well as the number of agents increases, but does not preserve equality with the strong actuated controller on congestion-efficiency metrics beyond the 2×2 network.

Table 4.4 shows how the learned-versus-Actuated gap changes with network size. This table complements the descriptive means by reporting the adjusted inferential evidence directly.

Table 4.4: Holm-adjusted learned-controller comparisons with Actuated control across the medium-demand scalability experiment.

Grid	Agents	Comparison	Completion Δ pp	Travel Δ s	Waiting Δ s	Time-loss Δ s	Queue Δ
2x2	4	DQN-Actuated	-0.293 (0.6508)	+1.86 (0.1255)	+0.64 (0.5984)	+1.90 (0.1284)	+0.064 (0.3559)
2x2	4	PPO-Actuated	-0.307 (0.6519)	+4.34 (0.0071*)	+2.22 (0.0061*)	+4.39 (0.0068*)	+0.383 (0.0055*)
3x3	9	DQN-Actuated	-0.244 (0.2316)	+3.78 (0.0019*)	+2.19 (0.0028*)	+3.79 (0.0015*)	+0.120 (0.0243*)
3x3	9	PPO-Actuated	-0.456 (0.0054*)	+8.43 (0.0003*)	+5.39 (0.0003*)	+8.44 (0.0003*)	+0.429 (0.0005*)
4x4	16	DQN-Actuated	-0.170 (0.3807)	+4.19 (0.0011*)	+2.88 (0.0005*)	+4.19 (0.0011*)	+0.095 (0.0014*)
4x4	16	PPO-Actuated	-0.650 (0.0144*)	+7.79 (0.0001*)	+5.43 (0.0001*)	+7.83 (0.0001*)	+0.266 (0.0001*)
5x5	25	DQN-Actuated	-0.243 (0.4244)	+4.06 (0.0054*)	+3.12 (0.0018*)	+4.11 (0.0045*)	+0.033 (0.0478*)
5x5	25	PPO-Actuated	-0.332 (0.2432)	+6.05 (0.0026*)	+4.57 (0.0011*)	+6.11 (0.0022*)	+0.084 (0.0064*)

Notes: Difference = learned controller minus Actuated. Positive completion differences favour the learned controller; negative travel, waiting, time-loss and queue differences favour the learned controller. Parentheses contain Holm-adjusted paired- t p -values; * denotes $p_H < 0.05$. Source: Author’s analysis of the frozen final medium-scalability paired-test outputs.

4.5.2 DQN versus PPO Across Scale

DQN and PPO had similar completion rates at four agents. From 9 agents onward, DQN produced significantly lower travel time, waiting time and queue length than PPO. The DQN-PPO completion difference was significant at 16 agents but not at 9 or 25 agents. This indicates that the clearest scaling advantage of DQN over PPO in this experiment concerns delay and queue efficiency rather than a uniform

throughput advantage.

Across the complete medium-demand scalability statistical analysis, 101 of 140 Holm-adjusted paired comparisons were significant.

4.6 Computational Scalability

Traffic performance does not by itself demonstrate that an independent-agent architecture remains computationally practical as the network grows. CPU/GPU utilisation percentages were not logged during the frozen runs and are not inferred retrospectively. Accordingly, Table 4.5 reports the measurements that were captured directly: training runtime, simulation/evaluation wall runtime, RAM, buffer memory, model size and inference latency.

Table 4.5: Measured computational scalability of independent DQN and PPO.

Method	Agents	Train (s)	Simulation wall time (s)	RAM (MB)	Buffer (MB)	Model (MB)	Latency (ms)
DQN	4	357.24	2.02	343.30	12.970	0.369	0.1978
DQN	9	692.93	3.96	357.94	29.182	0.829	0.3926
DQN	16	1046.14	6.84	388.56	51.880	1.475	0.6537
DQN	25	1398.06	10.63	422.66	81.062	2.304	1.0388
PPO	4	388.05	2.01	335.06	0.027	0.531	0.2827
PPO	9	731.54	4.05	334.73	0.062	1.195	0.5995
PPO	16	1018.70	6.95	339.75	0.109	2.125	0.9859
PPO	25	1469.70	10.79	343.27	0.171	3.320	1.5655

Source: Author’s measurements from the frozen final training and deterministic-evaluation pipeline.

Simulation wall time is the measured end-to-end wall-clock runtime of the 1,800-s deterministic simulation/evaluation run. The corresponding 95th-percentile joint decision latencies at 25 agents were 1.1955 ms for DQN and 1.8251 ms for PPO.

Figure 4.5 visualises the measured growth in training runtime. Training runtime increased from 357.24 s to 1,398.06 s for DQN and from 388.05 s to 1,469.70 s for PPO as agent count increased from 4 to 25. Descriptive least-squares fits over the four observed grid sizes gave slopes of 48.92 s per additional agent for DQN ($R^2 = 0.986$) and 50.08 s per agent for PPO ($R^2 = 0.993$). These fits summarise the observed range only and are not asymptotic complexity claims.

Total model storage increased linearly because each intersection owns an independent model. DQN model size rose from 0.369 MB at four agents to 2.304 MB at 25 agents, while PPO rose from 0.531 MB to 3.320 MB. Model size per agent remained effectively constant at approximately 0.092 MB for DQN and 0.133 MB for PPO.

DQN training RAM increased from 343.30 MB to 422.66 MB, whereas PPO training RAM remained comparatively stable, from 335.06 MB to 343.27 MB. DQN

replay-buffer memory grew from 12.97 MB to 81.06 MB; PPO rollout-buffer memory remained much smaller, from 0.027 MB to 0.171 MB.

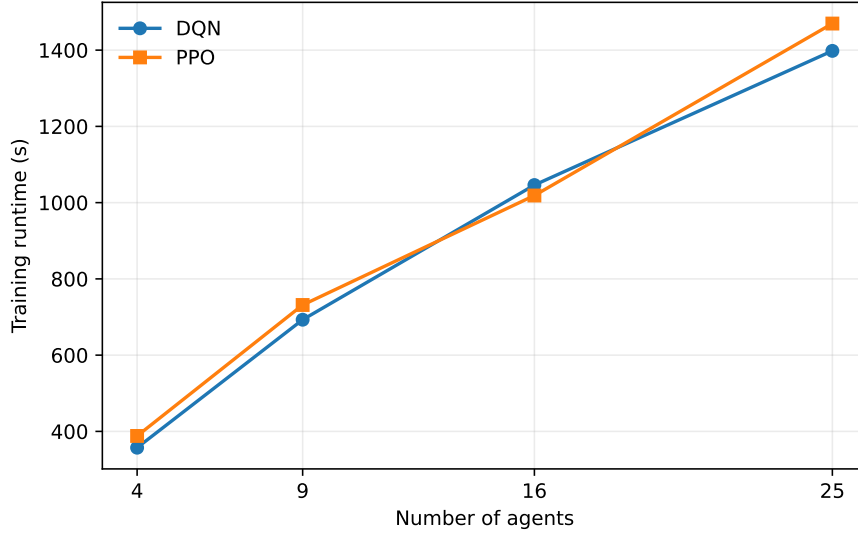


Figure 4.5: Training runtime versus number of independent agents.

Source: Author’s measurements from the final training pipeline.

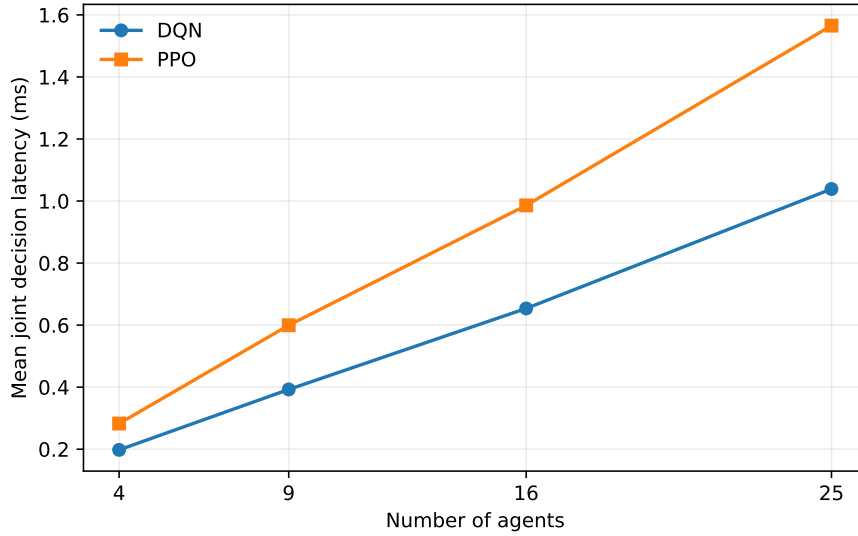


Figure 4.6: Mean deterministic joint decision latency versus number of independent agents.

Source: Author’s measurements from the final deterministic evaluation pipeline.

Figure 4.6 shows the growth in mean joint decision latency as agent count increases. Inference remained small relative to the five-second decision interval. At 25 agents, mean joint decision latency was 1.039 ms for DQN and 1.566 ms for PPO, corresponding to only 0.0208% and 0.0313% of the 5,000-ms control budget. The descriptive latency trend was almost linear over the four measured sizes ($R^2 = 0.999$ for both algorithms). Thus, within the tested 4–25-agent range, inference latency

was not the computational bottleneck; training time, cumulative model count and DQN replay memory grew more materially with system size.

4.7 Results in Relation to the Research Questions

The results provide direct evidence for the empirical research questions without requiring the stronger claim that MARL is universally superior.

For the demand-sensitivity question, controller ranking depended on congestion regime. DQN and PPO strongly outperformed the Original Fixed-Time controller under low, medium and dynamic demand, but the stronger Actuated controller was usually best descriptively. Under severe high demand, Webster was the strongest controller and PPO deteriorated sharply.

For the controller-effectiveness question, the answer therefore depends on baseline selection. DQN was competitive with strong conventional control under medium demand and preserved completion rate well across larger grids, but it did not outperform Actuated on congestion-efficiency metrics as scale increased. PPO showed weaker operational scalability.

For the scalability question, the generalised architecture successfully operated with 4, 9, 16 and 25 independent agents using the same fixed-dimension local observation and control interface. Behaviourally, DQN retained completion-rate performance but accumulated significant efficiency gaps relative to Actuated from 9 agents onward. Computationally, training and storage requirements increased with agent count, while joint deterministic inference remained far below the available decision-time budget.

4.8 Chapter Summary

This chapter reported the final 175-run evaluation using five controllers, four demand scenarios and four tested network sizes. The addition of Webster and Actuated baselines changes the interpretation of the original experiment: learned control provides large gains over the weak Original Fixed-Time programme, but strong conventional adaptive control remains highly competitive and is often superior.

Demand regime strongly affects algorithm ranking. DQN remained robust under the severe high-demand stress test, while PPO deteriorated substantially. Under calibrated medium demand, Actuated control produced the best descriptive traffic-efficiency means at all four network sizes. DQN preserved completion rate but became significantly less efficient than Actuated on travel time, waiting time and queue length from the 3×3 network onward. PPO scaled less favourably.

The computational analysis showed that model count, storage and training time increase with the number of independent agents, whereas deterministic inference remained extremely small relative to the five-second decision interval. These findings are interpreted critically in Chapter 5, with particular attention to baseline strength, saturation, independent-agent limitations and the distinction between architectural, behavioural and computational scalability.

Chapter 5 Discussion

5.1 Introduction

Chapter 4 showed that the final conclusions of this study depend strongly on three factors: the strength of the comparison baseline, the traffic-demand regime, and the distinction between architectural, behavioural and computational scalability. The revised experiment therefore produces a more qualified interpretation than a simple claim that Multi-Agent Reinforcement Learning (MARL) “outperforms fixed-time control”. DQN and PPO produced substantial improvements relative to the original 90-second fixed-time controller in several conditions, but the addition of Webster-derived and actuated baselines showed that a strong conventional controller can equal or outperform the learned policies.

This chapter interprets those results against the research questions and the traffic-signal-control literature. Particular attention is given to the implications of the strong actuated baseline, the severe high-demand stress test, the different scaling behaviour of DQN and PPO, and the computational cost of using one independent model per intersection. The discussion also separates what the experiment directly demonstrates from what remains outside the evidence.

5.2 Baseline Strength Changes the Interpretation of MARL Performance

One of the most important findings of the revised study is methodological rather than algorithmic: conclusions about reinforcement-learning performance change when the comparator changes. Against the Original Fixed-Time controller, both learned methods often appear strongly advantageous. Against Webster and especially Actuated control, the margin becomes much smaller and in several cases reverses.

This is consistent with the concern raised by Cabrejas-Egea et al. (2021), who noted that traffic-signal RL studies often lack comparison against strong operational control systems and therefore compared RL agents with commercial and conventional controllers. The final design of this thesis addresses the same evaluation problem by

retaining the original static baseline while adding a Webster-derived plan and an actuated controller.

The medium-demand 2×2 result is particularly informative. DQN achieved 94.13% completion compared with 94.43% for Actuated, and the Holm-adjusted paired difference was not statistically significant. DQN was also not significantly different from Actuated on travel time, waiting time, time loss or mean queue length. Therefore, under this condition DQN should be described as *competitive with* the strong conventional controller rather than superior to it.

PPO produced a weaker result. Although its completion rate remained similar, it had significantly larger travel time, waiting time, time loss and queue length than Actuated. Consequently, the revised evidence does not support treating DQN and PPO as interchangeable examples of “MARL performance”. Algorithm choice matters even when the state, action interface, reward structure, training budget and evaluation protocol are held broadly comparable.

This finding also supports the wider evaluation concerns documented by the systematic review of Noaeen et al. (2022). That review identified substantial variation in environments, evaluation practices and reproducibility across network-level RL traffic-signal studies. The present results show why this matters empirically: a controller that looks highly effective against one baseline can look merely competitive against another.

5.3 Demand Sensitivity and the Limits of Learned Control

5.3.1 Low and Medium Demand

Under low demand, all five controllers completed approximately 95% of planned trips, so the main differences concerned efficiency rather than throughput. Actuated control produced the lowest average travel time and waiting time, while DQN and PPO remained substantially better than the Original Fixed-Time controller. The result suggests that the 90-second static cycle was poorly matched to sparse arrivals, whereas adaptive controllers could avoid unnecessarily long red or green periods.

The medium-demand scenario provides the strongest evidence that DQN can operate effectively within the tested environment. DQN was statistically indistinguishable from Actuated on all five primary metrics after Holm–Bonferroni correction. This is important because it demonstrates that the independent DQN architecture can reach a level of traffic performance comparable to a strong demand-responsive conventional method without using a hand-designed actuated logic as its decision

policy.

However, the result should not be extrapolated into a general claim that learned control is preferable. Actuated remained descriptively better, and the experiment did not measure installation effort, detector calibration, controller certification, fault handling, explainability, maintenance cost or safety assurance. The comparison concerns simulated operational traffic metrics only.

5.3.2 Dynamic Demand

The dynamic scenario demonstrates adaptation to changing demand within an episode, but it does not establish out-of-distribution generalisation. The same route structure was part of the training configuration, while evaluation used different SUMO seeds. The correct interpretation is therefore that the learned policies could respond to temporal variation within a known scenario class.

DQN was again stronger than PPO, but Actuated achieved the best descriptive performance. This result is important because fluctuating demand is often used as a motivation for RL. The final experiment shows that temporal variability alone does not guarantee a learned controller will outperform a well-designed responsive conventional controller.

5.3.3 High-Demand Oversaturation

The high-demand scenario changes the interpretation most sharply. Even the strongest controller completed only approximately 15.5% of planned trips within the 1,800-second horizon. This indicates a network breakdown regime rather than normal congestion. The scenario should therefore be treated as an oversaturated stress test.

DQN remained comparatively robust: it was close to Webster on completion, travel time and waiting time and performed better than the weak Original Fixed-Time controller on key delay metrics. PPO, however, deteriorated substantially, with only 7.45% completion and much larger travel and waiting times. This divergence indicates that successful behaviour under low or medium demand is not sufficient evidence of robustness under severe saturation.

The result also reinforces the importance of evaluating several traffic regimes rather than reporting a single favourable demand condition. Noaeen et al. (2022) show that RL network-signal-control research is methodologically heterogeneous; the present experiment demonstrates the practical consequence of that heterogeneity because controller rankings are not invariant across demand regimes.

No causal claim is made about why PPO deteriorated. Its on-policy optimisation, exploration history, learned switching behaviour and interaction with the local reward

formulation are plausible mechanisms, but the final experiment was not designed to isolate them. Establishing causality would require controlled ablation experiments and additional training seeds.

5.4 DQN and PPO as Independent MARL Controllers

The final results favour DQN over PPO in the tested independent-agent configuration. Under the larger medium-demand networks, DQN consistently produced lower travel time, waiting time and queue length than PPO. Under high demand, the separation became substantially larger.

This does not imply that DQN is inherently superior to PPO for traffic signal control. The comparison is specific to the selected hyperparameters, reward function, binary action semantics, 50,000-step training budget and independent-agent architecture. PPO may require a different training horizon or policy configuration, while DQN benefits from replaying previous transitions and therefore uses experience differently.

The distinction between independent and explicitly cooperative MARL is also important. Cooperative traffic-signal methods frequently incorporate neighbour information, shared rewards, communication or centralised training mechanisms. Haddad et al. (2022), for example, explicitly model cooperation through neighbour state, action and reward information. Bie et al. (2024) use value decomposition and spatiotemporal graph attention to coordinate heterogeneous intersections. The present framework is intentionally simpler: each intersection owns an independent learner, receives a fixed-dimensional local observation containing only limited neighbour information, and has no shared critic or parameter-sharing mechanism.

That simplicity is a strength for implementation scalability, but it is also a likely limitation for coordination quality. As networks become larger, local actions can affect queues and arrivals several intersections away. A six-element local observation cannot represent the full joint traffic state. The performance gap between DQN and Actuated on delay metrics at larger grid sizes is therefore consistent with a broader MARL trade-off between decentralised scalability and coordination quality.

5.5 Behavioural Scalability

The framework successfully operated on 2×2 , 3×3 , 4×4 and 5×5 networks, corresponding to 4, 9, 16 and 25 independent agents. Intersection identifiers and neighbour relationships were discovered automatically, and the same six-dimensional

observation interface and two-action control design were used at every scale. This demonstrates *architectural scalability* across the tested range.

Behavioural scalability requires a more careful interpretation. DQN completion rate remained statistically indistinguishable from Actuated at all four tested network sizes. This suggests that DQN preserved the ability to serve a similar fraction of planned demand as the network grew.

At the same time, from nine agents onward DQN had significantly higher travel time, waiting time and mean queue length than Actuated. The traffic-efficiency gap therefore grew even though completion rate remained comparable. The experiment supports the statement that DQN preserves throughput-related behaviour better than delay-related efficiency, not the stronger statement that performance is invariant with network size.

PPO scaled less favourably. It showed significant delay and queue disadvantages relative to Actuated at every grid size and significant completion-rate disadvantages at the 3×3 and 4×4 sizes. DQN therefore exhibits the stronger behavioural scalability of the two learned controllers in this experiment.

The literature provides context for this trade-off. Chu et al. (2020) describe decentralised MARL as a response to the high-dimensional joint action problem of large-scale signal control, while also identifying partial observability and limited communication as challenges for local agents. The current results reflect the same tension: independent decentralisation makes the controller easy to instantiate at additional intersections, but increased network size does not remove the coordination problem.

Bie et al. (2024) further emphasise that realistic road networks can contain heterogeneous intersection geometry, demand and signal timing. The networks used in this thesis remain regular synthetic grids. Consequently, successful operation up to 25 agents demonstrates scalability of the implemented software architecture within homogeneous synthetic networks, but not robustness to the intersection heterogeneity found in real urban networks.

5.6 Computational Scalability

The computational results provide a second, separate meaning of scalability. At 25 agents, mean deterministic joint decision latency was approximately 1.04 ms for DQN and 1.57 ms for PPO. These values correspond to less than 0.032% of the 5,000-ms decision interval. Within the tested range, neural-network inference was therefore computationally negligible relative to the control interval.

Training cost increased more materially. DQN training runtime increased from

approximately 357 s at four agents to 1,398 s at 25 agents; PPO increased from approximately 388 s to 1,470 s. Saved model size also increased with agent count because every intersection owns a separate model. The descriptive linear trends observed over four grid sizes should be interpreted only within this measured range and not as proofs of asymptotic complexity.

DQN additionally incurred increasing replay-buffer memory, reaching approximately 81 MB at 25 agents, while PPO’s rollout-buffer footprint remained much smaller. DQN training RAM nevertheless remained below approximately 423 MB in the measured experiments, and PPO remained below approximately 344 MB.

These results support a limited but useful conclusion: the independent-agent architecture is computationally feasible for deterministic inference across the tested 4–25-agent range. CPU/GPU utilisation percentages were not logged during the frozen runs; consequently, the computational conclusions are based on measured wall time, memory, storage and inference latency rather than reconstructed processor-utilisation estimates. They do not demonstrate deployment readiness. Real systems would add sensor processing, communication delay, controller interfaces, redundancy, monitoring, security and potentially edge-hardware constraints that are absent from the current measurement.

5.7 Implications for Adaptive Traffic Signal Control

5.7.1 Theoretical Implications

The study illustrates that scalability in MARL is multidimensional. A controller can scale architecturally because the same code instantiates more agents, computationally because inference remains within the real-time decision budget, and yet lose relative behavioural efficiency as coordination becomes harder. These dimensions should therefore be reported separately.

The results also support the importance of local-state design and reward construction. The independent agents were able to maintain high completion rates using only compact local observations, indicating that a small state representation can retain useful control information. However, the growing delay gap relative to Actuated suggests that compact local state alone may be insufficient for network-level coordination at larger scales.

A future extension could therefore test whether explicit communication, graph-based neighbour encoding, parameter sharing, value decomposition or centralised training with decentralised execution improves the delay-efficiency gap. Such extensions would connect the present independent-agent baseline with more explicitly cooperative approaches such as those investigated by Haddad et al. (2022) and Bie

et al. (2024).

5.7.2 Methodological Implications

The revised experiment demonstrates why traffic-control studies should use multiple baselines and multiple demand regimes. A conclusion based only on Original Fixed-Time control would have substantially overstated the advantage of MARL. Adding Webster and Actuated control produced a more demanding test and changed the final conclusion from “MARL is superior” to “DQN is competitive in some conditions, but strong conventional adaptive control remains difficult to beat.”

The study also demonstrates the importance of multiplicity control. Seven predefined controller comparisons were conducted for each metric family, and Holm correction was applied before inferential claims were made. This is more conservative than interpreting a large collection of unadjusted p -values independently.

Finally, the experiment separates descriptive differences from inferential conclusions. Small numerical differences, such as the 25-agent DQN–Actuated queue difference, can be statistically detectable while remaining small in practical magnitude. Conversely, five paired seeds provide limited statistical resolution. Effect size, confidence interval, direction and domain meaning must therefore accompany significance testing.

5.7.3 Practical Implications

The results do not justify replacing actuated traffic controllers with DQN or PPO. Instead, they suggest that independent DQN is a plausible research baseline for adaptive control because it achieved competitive medium-demand completion and very low inference latency while using a simple decentralised architecture.

For practitioners, the stronger lesson is evaluation discipline. Any proposed learning controller should be compared against a realistic conventional method, evaluated across several demand regimes, and assessed on several operational metrics rather than reward alone. Cabrejas-Egea et al. (2021) similarly motivate comparison with stronger operational systems rather than relying only on weak cyclic baselines.

5.8 Research Questions Revisited

5.8.1 RQ1: How can an adaptive urban traffic signal control system based on a scalable independent MARL approach be designed and implemented using Python and SUMO?

The study answers RQ1 through the implemented architecture. Each SUMO traffic-light intersection is automatically discovered and represented by an independent DQN or PPO agent. The same six-element observation, binary HOLD/SWITCH action interface, local reward structure and five-second decision interval are reused across networks from four to twenty-five intersections. This demonstrates that the software architecture can scale to additional grid intersections without hard-coded agent identifiers.

5.8.2 RQ2: How effective are independent DQN and PPO traffic-signal controllers under low, medium, high and dynamic traffic-demand conditions?

Effectiveness depends on demand. DQN and PPO substantially improved delay-based metrics relative to Original Fixed-Time control under low, medium and dynamic demand, but Actuated control was generally stronger. Under the high-demand oversaturated stress test, DQN remained comparatively robust while PPO deteriorated substantially. The framework is therefore demand-sensitive rather than uniformly effective.

5.8.3 RQ3: How well does MARL-based control improve trip completion, waiting time, queue length, travel time and time loss compared with conventional fixed-time and actuated traffic signal control?

The answer depends on which conventional controller is used. Relative to the Original Fixed-Time programme, the learned methods often produced large improvements. Relative to Webster and especially Actuated control, the advantage was much smaller or absent. DQN was competitive with Actuated on the 2×2 medium-demand network and preserved similar completion rates across scale, but Actuated achieved significantly lower delay and queue metrics on the larger networks. The final evidence therefore does not support universal MARL superiority.

5.8.4 RQ4: How can a reproducible Data Science workflow be used to assess the behavioural and computational scalability of the MARL framework as the number of controlled intersections increases?

The study used fixed scenario definitions, matched evaluation seeds, retained raw SUMO outputs, separate derived analysis files, predefined paired comparisons, confidence intervals, paired effect sizes, Holm–Bonferroni multiplicity correction and computational-resource measurements. Frozen checksummed evidence was used to generate the final thesis tables and figures. This workflow allowed favourable, neutral and unfavourable outcomes to be reported consistently and made the final conclusions traceable to retained numerical evidence.

5.9 Limitations and Threats to Interpretation

Several limitations bound the conclusions.

First, all networks are synthetic and predominantly regular grids. Real cities contain heterogeneous junction geometry, uneven lane capacities, multimodal traffic, incidents and irregular turning patterns. The study therefore demonstrates simulation-scale behaviour, not field performance.

Second, the larger-grid evaluation uses only calibrated medium demand. The 2×2 network was tested under four demand regimes, but 3×3 , 4×4 and 5×5 networks were not subjected to equivalent low, high and dynamic demand experiments. Demand robustness and network-size robustness are therefore only partially crossed.

Third, each learned configuration was trained with a single training seed. Five matched evaluation seeds provide repeated performance observations, but they do not quantify variability caused by different neural-network initialisations or training trajectories.

Fourth, the evaluation sample contains only five matched seeds per comparison. Holm correction controls family-wise multiplicity, but confidence intervals and paired t -tests remain sensitive to the small sample. The exact sign-flip test has a minimum two-sided resolution of 0.0625 at $n = 5$ and is therefore useful only as a sensitivity analysis.

Fifth, the independent-agent architecture has no explicit communication, shared critic or parameter sharing. This simplifies implementation but limits the information available for coordination across distant intersections.

Sixth, the high-demand scenario is a severe oversaturated stress test. Its very low completion rates mean that travel-time and waiting-time averages are calculated

from the subset of trips that finish within the simulation horizon. They should not be interpreted as normal high-congestion operating conditions.

Seventh, computational measurements are hardware- and implementation-specific. They demonstrate feasibility on the machine and software stack used for this thesis, not universal latency or memory requirements.

Finally, simulation omits several deployment factors, including detector noise, communication failures, pedestrian phases, emergency priority, field-controller safety logic and institutional constraints. These omissions prevent any claim of immediate real-world deployment readiness.

5.10 Chapter Summary

The revised findings lead to a deliberately qualified conclusion. Independent MARL is technically scalable across the tested 4–25-agent synthetic networks and computationally lightweight at inference time. DQN is the stronger learned controller in this experiment and preserves completion rate well as network size increases. However, strong conventional adaptive control remains highly competitive: Actuated control achieved the best descriptive medium-demand traffic performance at every tested grid size and developed significant efficiency advantages over DQN from nine agents onward.

Demand regime also matters. Low, medium and dynamic conditions show clear advantages over the weak Original Fixed-Time baseline, while the high-demand stress test exposes substantial fragility, especially for PPO. These results reinforce the need for strong baselines, multiple traffic regimes, corrected statistical inference and explicit separation of architectural, behavioural and computational scalability.

Chapter 6 consolidates these findings, states the final contributions of the research, and presents recommendations for future work.

Chapter 6 Conclusion

6.1 Introduction

This thesis investigated whether an independent Multi-Agent Reinforcement Learning (MARL) architecture can provide adaptive and scalable traffic signal control in synthetic urban road networks. The study implemented DQN- and PPO-based traffic-signal agents in Python and SUMO, evaluated them against three conventional controllers, tested demand sensitivity on a primary 2×2 network, extended the same architecture to grids containing up to 25 signalised intersections, and measured both traffic and computational scalability.

The final experiment consisted of 175 unique deterministic evaluation runs using five matched evaluation seeds. The revised evidence shows that MARL performance is highly dependent on baseline strength, congestion regime and algorithm choice. DQN emerged as the stronger learned controller, but the Actuated controller remained the strongest conventional benchmark under calibrated medium-demand conditions. The main contribution of the study is therefore not a claim that MARL universally outperforms conventional control, but a reproducible evaluation showing where independent MARL is competitive, where it loses efficiency, and how its computational cost changes as the number of agents increases.

6.2 Summary of the Study

The research began from the problem that static traffic-signal plans are unable to react directly to continuously changing local traffic conditions. Reinforcement learning offers an alternative in which controllers learn signal decisions from interaction with a simulated traffic environment. When several intersections must be controlled together, MARL provides a distributed formulation in which each intersection acts as an agent.

A scalable independent-agent framework was implemented using SUMO and TraCI. Traffic-light agents and their neighbours were discovered automatically from network topology. Each agent used a fixed six-element local observation, a two-action

HOLD/SWITCH control interface, a minimum green duration of 10 seconds, a three-second yellow transition and a five-second decision interval. Separate DQN and PPO models were trained for every controlled intersection.

The final experimental design compared five controllers:

- Original Fixed-Time;
- Webster-derived Fixed-Time;
- Actuated;
- independent DQN;
- independent PPO.

The primary 2×2 demand experiment evaluated low, medium, high and dynamic traffic. A second experiment evaluated medium-demand scalability on 2×2 , 3×3 , 4×4 and 5×5 networks containing 4, 9, 16 and 25 agents. Learned models were trained using a fixed 50,000-step budget and training seed 1 and evaluated using matched seeds 11–15.

Final analysis used trip completion rate, average travel time, average waiting time, average time loss and mean sampled queue length. Statistical comparisons used paired tests with Holm–Bonferroni correction for the predefined comparison families, supplemented by confidence intervals, paired effect sizes and an exact sign-flip sensitivity analysis. Computational scalability was assessed using training runtime, training RAM, model size, buffer memory and deterministic joint decision latency.

6.3 Answers to the Research Questions

6.3.1 RQ1: How can an adaptive urban traffic signal control system based on a scalable independent MARL approach be designed and implemented using Python and SUMO?

The study demonstrated that a scalable MARL traffic-control architecture can be implemented by representing every signalised intersection as an independent learning agent connected to SUMO through TraCI. Rather than hard-coding intersection identifiers, the framework discovers controlled intersections and direct neighbours from each SUMO network.

The same local state structure, reward logic and binary HOLD/SWITCH action interface were applied without structural modification to networks containing 4, 9, 16 and 25 controlled intersections. This provides direct evidence of architectural scalability across the tested grid sizes.

The corrected final environment also demonstrated the importance of explicit signal semantics. Action 0 holds the current green rather than allowing the underlying static programme to continue automatically, while Action 1 requests a legal transition after the minimum-green constraint. Retraining under these final semantics was necessary because learned behaviour is dependent on the action implementation.

6.3.2 RQ2: How effective are independent DQN and PPO traffic-signal controllers under low, medium, high and dynamic traffic-demand conditions?

The effectiveness of MARL varied substantially by demand regime.

Under low demand, both DQN and PPO improved delay-related metrics relative to the Original Fixed-Time controller, although Actuated control remained descriptively best.

Under medium demand, DQN was the strongest learned controller and was statistically indistinguishable from Actuated on completion, travel time, waiting time, time loss and queue length on the 2×2 network after Holm–Bonferroni correction. PPO was weaker and showed statistically higher delay and queue metrics than Actuated.

Under dynamic demand, DQN again adapted effectively relative to the Original Fixed-Time controller but remained weaker than Actuated. The result demonstrates adaptation to temporal demand variation within a trained scenario class, not out-of-distribution generalisation.

Under high demand, the network entered severe oversaturation. Even the strongest controller completed only approximately 15.5% of planned trips within the simulation horizon. DQN remained comparatively robust, while PPO performance deteriorated sharply. This shows that performance under light or moderate traffic cannot be assumed to transfer to severe breakdown conditions.

The answer to RQ2 is therefore conditional: independent MARL is effective in several low-to-medium and changing-demand conditions, but its effectiveness is not uniform across congestion regimes.

6.3.3 RQ3: How well does MARL-based control improve trip completion, waiting time, queue length, travel time and time loss compared with conventional fixed-time and actuated traffic signal control?

The final results show that the answer depends on the conventional baseline.

Against the Original Fixed-Time controller, DQN and PPO often produced large reductions in travel time, waiting time and queue length. However, the addition of Webster and Actuated controllers substantially changed the interpretation.

Under medium demand, Actuated control produced the best descriptive completion rate, travel time, waiting time and queue length at every tested grid size. DQN retained completion performance close to Actuated, and completion differences were not statistically significant at 4, 9, 16 or 25 agents. Nevertheless, from the 3×3 network onward DQN showed significantly larger travel time, waiting time and queue length than Actuated.

PPO scaled less favourably and was generally weaker than DQN on delay and queue metrics. It also failed substantially under the severe high-demand stress test.

The final answer to RQ3 is therefore that MARL can strongly outperform a weak static baseline, but it does not universally outperform strong conventional adaptive control. DQN is competitive in some conditions and preserves throughput-related performance well, whereas Actuated remains the most efficient medium-demand controller in the final experiment.

6.3.4 RQ4: How can a reproducible Data Science workflow be used to assess the behavioural and computational scalability of the MARL framework as the number of controlled intersections increases?

The study implemented a reproducible workflow covering network and demand generation, model training, deterministic evaluation, collection of SUMO outputs, descriptive analysis, matched statistical comparison, computational profiling and thesis-ready visualisation.

The final analysis used consistent metrics across controllers, matched evaluation seeds, predefined pairwise comparisons, Holm–Bonferroni multiplicity correction, confidence intervals, paired effect sizes and exact sign-flip sensitivity checks. Raw and derived results were retained separately, and the final numerical evidence was frozen using checksums before thesis tables and figures were generated.

This workflow made it possible to report favourable, neutral and unfavourable

results using the same analytical rules. It also allowed behavioural and computational scalability to be evaluated separately rather than treating successful execution on a larger network as sufficient evidence of scalable performance.

6.4 Main Contributions of the Research

The thesis makes five principal contributions.

First, it implements a generalised independent-agent MARL traffic-signal-control architecture that automatically instantiates agents and neighbour relationships from SUMO network topology. The same implementation operates across the tested range from 4 to 25 controlled intersections.

Second, it provides a corrected and explicitly validated traffic-signal action model based on HOLD and SWITCH decisions with enforceable signal-safety timing constraints. This avoids ambiguity between agent decisions and the simulator’s underlying signal programme.

Third, the study uses stronger conventional control baselines than the initial fixed-time-only comparison. Including Webster and Actuated controllers demonstrates that baseline strength materially changes the apparent benefit of MARL.

Fourth, the study evaluates scalability using both traffic behaviour and computational measurements. DQN preserves completion-rate performance comparatively well across the tested range, while inference latency remains extremely small relative to the five-second control budget. At the same time, training time, replay memory and total model size increase with agent count.

Fifth, the study applies a reproducible Data Science evaluation protocol with matched seeds, multiple traffic metrics, Holm-adjusted paired comparisons, effect-size reporting and frozen evidence. This improves the transparency of the conclusions and reduces the risk of relying on isolated favourable simulations.

6.5 Key Findings

The final findings can be summarised as follows.

- Actuated control was the strongest descriptive controller under calibrated medium demand at all four tested grid sizes.
- DQN was the stronger learned controller and was competitive with Actuated on the primary 2×2 medium-demand network.
- DQN preserved completion rate across larger grids, but from nine agents onward showed significant efficiency disadvantages relative to Actuated on travel time,

waiting time and queue length.

- PPO was generally weaker than DQN and deteriorated substantially under severe high-demand oversaturation.
- Webster was especially strong in the oversaturated high-demand stress test.
- The same MARL software architecture successfully instantiated 4, 9, 16 and 25 independent agents without hard-coded network-specific control logic.
- Deterministic inference remained computationally lightweight: at 25 agents, joint mean decision latency was approximately 1.04 ms for DQN and 1.57 ms for PPO, far below the 5,000-ms decision interval.
- Training time, total model storage and DQN replay-buffer memory increased as the number of agents increased.
- Stronger baselines and multiplicity correction materially changed the final interpretation compared with a simple fixed-time-only evaluation.

6.6 Limitations

The conclusions of this study are bounded by several limitations.

The networks are synthetic regular grids rather than real urban road networks. Intersection geometry, turning behaviour, lane counts and demand structure are therefore less heterogeneous than in operational traffic systems.

The larger-grid scalability experiment was conducted under calibrated medium demand only. Low, high and dynamic traffic were not repeated across every larger network size. Consequently, demand robustness and network-size robustness were not fully crossed in a single factorial experiment.

Each learned configuration used one training seed. Although five matched evaluation seeds were used, the study does not quantify training variability across multiple neural-network initialisations and learning trajectories.

Five evaluation seeds also remain a small inferential sample. Holm–Bonferroni correction reduces family-wise false-positive risk, but confidence intervals and paired tests remain sensitive to small-sample variation.

The independent-agent architecture contains no explicit communication, shared critic or parameter sharing. Local observations may therefore be insufficient to capture long-range traffic interactions as the network grows.

The severe high-demand experiment represents a breakdown condition. Because only completed trips contribute to several trip-level means, high-demand travel and waiting times must be interpreted alongside the very low trip-completion rate.

Computational measurements are specific to the hardware, software environment, implementation and network sizes used for the study.

Finally, SUMO does not reproduce every field-deployment constraint. Detector noise, communication loss, pedestrian calls, emergency priority, safety certification, hardware faults and institutional operating procedures remain outside the experiment.

6.7 Recommendations for Future Research

Future research should first evaluate the framework across more training seeds. This would quantify the sensitivity of DQN and PPO to neural-network initialisation and learning stochasticity rather than measuring evaluation variability alone.

Second, the larger-grid networks should be tested under low, high and dynamic demand in addition to medium demand. This would allow demand robustness and network scalability to be evaluated jointly.

Third, future work should compare the independent architecture with explicitly cooperative MARL methods, including parameter sharing, centralised training with decentralised execution, graph neural networks, value decomposition or neighbour communication. This would test whether additional coordination reduces the growing delay gap observed against Actuated control.

Fourth, more heterogeneous networks should be introduced. Irregular junctions, unequal approach capacities, asymmetric turning flows and realistic route structures would provide a stronger test of generalisation.

Fifth, robustness should be evaluated under detector noise, delayed observations, communication failure and incidents. Such experiments would narrow the gap between idealised simulation and deployable intelligent transportation systems.

Sixth, the reward function should be investigated through controlled ablation experiments. Individual terms for waiting time, queue, discharge and switching could be removed or reweighted to determine their causal contribution to learned behaviour, particularly under oversaturation.

Seventh, future evaluation should include additional objectives such as emissions, fuel consumption, pedestrian delay, fairness between approaches, emergency-vehicle priority and public-transport priority.

Finally, a realistic progression toward deployment would require hardware-in-the-loop or controller-in-the-loop evaluation before any field trial. Such work should include formal signal-safety constraints, fail-safe fallback control and operational monitoring.

6.8 Final Conclusion

This thesis demonstrates that the independent MARL implementation can be generalised across the tested 2×2 to 5×5 synthetic networks, corresponding to four to twenty-five controlled intersections, while keeping a fixed local observation and control interface. Deterministic inference remained far below the available decision-time budget throughout this measured range. These findings constitute evidence of architectural and computational scaling over the tested 4–25-agent range; they are not a claim of unbounded or city-scale scalability.

However, the final evidence also shows why claims of MARL superiority must be made cautiously. When only the Original Fixed-Time controller is used as a baseline, the learned controllers appear substantially better. When Webster and Actuated control are added, the comparison becomes more demanding. Actuated control is the strongest medium-demand controller in the final experiment, while DQN remains competitive on completion rate but loses delay efficiency as the network grows. PPO performs less consistently and is particularly vulnerable under severe oversaturation.

The principal conclusion is therefore not that reinforcement learning should replace conventional adaptive signal control. Rather, independent DQN provides a viable research approach whose computational behaviour remains practical across the tested 4–25-agent range and whose strengths and limitations can be measured systematically. Strong baselines, varied demand conditions, reproducible statistical analysis and explicit computational measurement are essential if future MARL traffic-signal research is to move from algorithm demonstrations toward credible evidence for intelligent transportation systems.

Bibliography

- Alvarez Lopez, P., Behrisch, M., Bieker-Walz, L., Erdmann, J., Flötteröd, Y.-P., Hilbrich, R., Lücken, L., Rummel, J., Wagner, P. & Wießner, E. (2018), Microscopic traffic simulation using sumo, *in* ‘2018 21st International Conference on Intelligent Transportation Systems (ITSC)’, IEEE, pp. 2575–2582.
- Bie, Y., Ji, Y. & Ma, D. (2024), ‘Multi-agent deep reinforcement learning collaborative traffic signal control method considering intersection heterogeneity’, *Transportation Research Part C: Emerging Technologies* **164**, 104663.
- Bouktif, S., Cheniki, A., Ouni, A. & El-Sayed, H. (2023), ‘Deep reinforcement learning for traffic signal control with consistent state and reward design approach’, *Knowledge-Based Systems* **267**, 110440.
- Cabrejas-Egea, A., Zhang, R. & Walton, N. (2021), ‘Reinforcement learning for traffic signal control: Comparison with commercial systems’, *Transportation Research Procedia* **58**, 638–645.
- Cai, S., Fang, J. & Xu, M. (2025), ‘Xlight: An interpretable multi-agent reinforcement learning approach for traffic signal control’, *Expert Systems with Applications* **273**, 126938.
- Chen, C., Wei, H., Xu, N., Zheng, G., Yang, M., Xiong, Y., Xu, K. & Li, Z. (2020), Toward a thousand lights: Decentralized deep reinforcement learning for large-scale traffic signal control, *in* ‘Proceedings of the AAAI Conference on Artificial Intelligence’, Vol. 34, pp. 3414–3421.
- Chu, T., Wang, J., Codecà, L. & Li, Z. (2020), ‘Multi-agent deep reinforcement learning for large-scale traffic signal control’, *IEEE Transactions on Intelligent Transportation Systems* **21**(3), 1086–1095.
- Cohen, J. (1988), *Statistical Power Analysis for the Behavioral Sciences*, 2 edn, Lawrence Erlbaum Associates.

- Du, W., Ye, J., Gu, J., Li, J., Wei, H. & Wang, G. (2023), Safelight: A reinforcement learning method toward collision-free traffic signal control, *in* ‘Proceedings of the AAAI Conference on Artificial Intelligence’, Vol. 37, pp. 14801–14810.
- Eclipse SUMO (2026a), ‘Traci traffic-light control – sumo documentation’, Official SUMO documentation. Accessed 17 August 2026, https://sumo.dlr.de/docs/TraCI/Change_Traffic_Lights_State.html.
- Eclipse SUMO (2026b), ‘Traffic light tools – sumo documentation’, Official SUMO documentation. Accessed 17 August 2026, <https://sumo.dlr.de/docs/Tools/tls.html>.
- Eclipse SUMO (2026c), ‘Traffic lights – sumo documentation’, Official SUMO documentation. Accessed 17 August 2026, https://sumo.dlr.de/docs/Simulation/Traffic_Lights.html.
- Federal Highway Administration (2008), Traffic signal timing manual, Technical Report FHWA-HOP-08-024, U.S. Department of Transportation, Federal Highway Administration.
- Gu, Y., Zhang, K., Liu, Q., Gao, W., Li, L. & Zhou, J. (2024), π -light: Programmatic interpretable reinforcement learning for resource-limited traffic signal control, *in* ‘Proceedings of the AAAI Conference on Artificial Intelligence’, Vol. 38, pp. 21107–21115.
- Haddad, T. A., Hedjazi, D. & Aouag, S. (2022), ‘A deep reinforcement learning-based cooperative approach for multi-intersection traffic signal control’, *Engineering Applications of Artificial Intelligence* **114**, 105019.
- Holm, S. (1979), ‘A simple sequentially rejective multiple test procedure’, *Scandinavian Journal of Statistics* **6**(2), 65–70.
- Katzilieris, K., Kampitakis, E. & Vlahogianni, E. I. (2026), ‘A multi-agent reinforcement learning framework for integrated traffic signal control and dynamic bus lane access management’, *Transportation Research Part C: Emerging Technologies* **182**, 105391.
- Kuyer, L., Whiteson, S., Bakker, B. & Vlassis, N. (2008), Multiagent reinforcement learning for urban traffic control using coordination graphs, *in* ‘Machine Learning and Knowledge Discovery in Databases: ECML PKDD 2008’, Vol. 5211 of *Lecture Notes in Computer Science*, Springer, pp. 656–671.
- Lakens, D. (2013), ‘Calculating and reporting effect sizes to facilitate cumulative science: A practical primer for t-tests and anovas’, *Frontiers in Psychology* **4**, 863.

- Liu, Y., Luo, G., Yuan, Q., Li, J., Jin, L., Chen, B. & Pan, R. (2023), Gplight: Grouped multi-agent reinforcement learning for large-scale traffic signal control, *in* ‘Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence’, pp. 199–207.
- Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., Graves, A., Riedmiller, M., Fidjeland, A. K., Ostrovski, G., Petersen, S., Beattie, C., Sadik, A., Antonoglou, I., King, H., Kumaran, D., Wierstra, D., Legg, S. & Hassabis, D. (2015), ‘Human-level control through deep reinforcement learning’, *Nature* **518**, 529–533.
- Noaeen, M., Naik, A., Goodman, L., Crebo, J., Abrar, T., Shakeri Hossein Abad, Z., Bazzan, A. L. C. & Far, B. (2022), ‘Reinforcement learning in urban network traffic signal control: A systematic literature review’, *Expert Systems with Applications* **199**, 116830.
- Raffin, A., Hill, A., Gleave, A., Kanervisto, A., Ernestus, M. & Dormann, N. (2021), ‘Stable-baselines3: Reliable reinforcement learning implementations’, *Journal of Machine Learning Research* **22**(268), 1–8.
- Ren, Y., Chang, Y., Cui, Z., Chang, X., Yu, H., Li, X. & Wang, Y. (2025), ‘Is cooperative always better? multi-agent reinforcement learning with explicit neighborhood backtracking for network-wide traffic signal control’, *Transportation Research Part C: Emerging Technologies* **179**, 105265.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A. & Klimov, O. (2017), ‘Proximal policy optimization algorithms’, arXiv:1707.06347.
- Song, X. B., Zhou, B. & Ma, D. (2024), ‘Cooperative traffic signal control through a counterfactual multi-agent deep actor critic approach’, *Transportation Research Part C: Emerging Technologies* **160**, 104528.
- Sutton, R. S. & Barto, A. G. (2018), *Reinforcement Learning: An Introduction*, 2 edn, MIT Press.
- Webster, F. V. (1958), Traffic signal settings, Technical Report Technical Paper No. 39, Road Research Laboratory.
- Wei, H., Chen, C., Zheng, G., Wu, K., Gayah, V., Xu, K. & Li, Z. (2019), Presslight: Learning max pressure control to coordinate traffic signals in arterial network, *in* ‘Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining’, Association for Computing Machinery, pp. 1290–1298.

- Wei, H., Xu, N., Zhang, H., Zheng, G., Zang, X., Chen, C., Zhang, W., Zhu, Y., Xu, K. & Li, Z. (2019), Colight: Learning network-level cooperation for traffic signal control, *in* ‘Proceedings of the 28th ACM International Conference on Information and Knowledge Management’, Association for Computing Machinery, pp. 1913–1922.
- Wei, H., Zheng, G., Gayah, V. & Li, Z. (2021), ‘Recent advances in reinforcement learning for traffic signal control: A survey of models and evaluation’, *ACM SIGKDD Explorations Newsletter* **22**(2), 12–18.
- Wei, H., Zheng, G., Yao, H. & Li, Z. (2018), Intellilight: A reinforcement learning approach for intelligent traffic light control, *in* ‘Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining’, Association for Computing Machinery, pp. 2496–2505.
- Wu, Q., Wu, J., Shen, J., Du, B., Telikani, A., Fahmideh, M. & Liang, C. (2022), ‘Distributed agent-based deep reinforcement learning for large scale traffic signal control’, *Knowledge-Based Systems* **241**, 108304.
- Xing, D., Zheng, Q., Liu, Q. & Pan, G. (2022), Tinylight: Adaptive traffic signal control on devices with extremely limited resources, *in* ‘Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence’, pp. 3999–4005.
- Zhang, H., Feng, S., Liu, C., Ding, Y., Zhu, Y., Zhou, Z., Zhang, W., Yu, Y., Jin, H. & Li, Z. (2019), Cityflow: A multi-agent reinforcement learning environment for large scale city traffic scenario, *in* ‘The World Wide Web Conference’, Association for Computing Machinery, pp. 3620–3624.
- Zheng, G., Xiong, Y., Zang, X., Feng, J., Wei, H., Zhang, H., Li, Y., Xu, K. & Li, Z. (2019), Learning phase competition for traffic signal control, *in* ‘Proceedings of the 28th ACM International Conference on Information and Knowledge Management’, Association for Computing Machinery, pp. 1963–1972.

Appendix A Reproducibility Details

This appendix records the final experimental settings needed to interpret or reproduce the submitted final evidence. It complements the methodological description in Chapter 3.

A.1 Core Simulation Settings

All final controller evaluations used a simulation horizon of 1,800 s. The MARL decision interval was 5 s. The corrected traffic-signal environment enforced a 10 s minimum green and a 3 s yellow transition. Evaluation seeds were 11–15 for every controller within a matched network–demand configuration. Learned-controller training used training seed 1 and a fixed budget of 50,000 decision steps per algorithm–grid–scenario training job.

The final evaluation contains 175 unique runs: 100 runs for the 2×2 demand experiment and 75 additional medium-demand runs for the 3×3 , 4×4 and 5×5 scalability extension.

A.2 Evidence Retention

The retained evidence includes network and route files, final DQN and PPO models, training logs, step-level evaluation data files, SUMO tripinfo output files, conventional-controller summaries, learned-controller summaries, statistical-analysis outputs and computational-resource summaries.

After the final analysis pipeline was completed, the authoritative numerical inputs used for thesis reporting were frozen using checksum verification. Thesis tables and figures were generated from these retained analysis files rather than by manual transcription.

A.3 Statistical Reproducibility

Controller comparisons use matched seeds and predefined paired comparisons. Holm–Bonferroni step-down correction is applied within each seven-comparison scenario–metric or grid–metric family. Reported inferential outputs include the paired mean difference, 95% confidence interval, paired-samples Cohen’s d_z , median paired difference and an exact two-sided sign-flip sensitivity result.

Because only five matched seed pairs are available, the exact two-sided sign-flip procedure has a minimum attainable probability of 0.0625. It is therefore retained as a sensitivity analysis rather than used as the sole decision rule.

Appendix B Code and Data Availability

The implementation associated with this thesis is maintained in the project repository:

`https://github.com/Venushambhu/
Scalable-MARL-for-adaptive-traffic-signal-control-`

The project contains the Python–SUMO–TraCI environment and the scripts used for network/scenario generation, DQN and PPO training, controller evaluation and analysis. The final thesis revision used a separated final-evaluation workflow so that corrected experiments and derived results did not overwrite the original development outputs.

The final numerical claims in this dissertation were generated from retained final result files and checksum-verified analysis inputs. Where complete model files or large raw simulation artefacts are not stored directly in the public repository, the thesis reports the configuration, seeds, metrics and analysis procedure required to identify the corresponding final evidence.